

Take the Short Route in Confidence Crises*

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Abstract

During the 2010–2012 European debt crisis, Italian spreads spiked while debt maturity shortened. Existing quantitative work attributes only a modest role to self-fulfilling risk. We develop a sovereign debt model with endogenous maturity and Calvo-style self-fulfilling crises in which long-term bonds carry within-period confidence risk and short-term debt insures against it. Maturity shortening is the optimal response to the increase in confidence risk, not evidence against it. A particle filter on Italian data attributes roughly 80 percent of the late-2011 spread spike to a rise in confidence risk.

Keywords: Sovereign debt crises, self-fulfilling crises, debt maturity, confidence risk

JEL classification: F34, H63, G12, E62

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1 Introduction

During the European sovereign debt crisis of 2010–2012, Italian sovereign spreads spiked, peaking above 500 basis points in late 2011. This rise prompted an active policy debate about its source. European policymakers argued that the spike could not be reconciled with the underlying macroeconomic deterioration, which was modest by comparison: output contracted during the 2008–2009 and 2011–2012 recessions, and debt-to-GDP rose by roughly 20 percentage points. They instead invoked a narrative of a “confidence crisis” to justify their interventions, arguing that pessimistic investor beliefs had become self-reinforcing: nervous investors demanded high interest rates, raising debt payments, compromising fiscal sustainability, and in turn reinforcing investors’ initial fears. However, the existing quantitative work on this episode cannot reproduce the spike. Using the canonical Cole–Kehoe framework with endogenous maturity, calibrated to Italian data, [Bocola and Dovis \(2019\)](#) argue that the government’s maturity choice is informative about changes in confidence risk. Disciplining the framework with the maturity choice of the government, they attribute only 13 percent of the spread to confidence risk, so the model cannot account for the observed spike in spreads.

Theoretical work points to a different source of multiplicity in sovereign debt markets, one that operates through long-term bonds. [Lorenzoni and Werning \(2019\)](#) and [Aguiar and Amador \(2020\)](#) show that the interaction between the long-term bond price and new issuance generates self-fulfilling multiplicity. In this paper, we propose a Calvo-style framework in which Italy’s decision to shorten debt maturity during the crisis is the optimal response to rising confidence risk associated with long-term bonds. As a result, short-term debt reduces the government’s exposure to a confidence crisis in the short run, so a rising probability of a crisis tilts the optimal portfolio toward shorter bonds. Our main result is that self-fulfilling crises account for approximately 80 percent of the Italian spread at the late-2011 peak.

Our theory builds on [Lorenzoni and Werning \(2019\)](#), a Calvo-style framework in which self-fulfilling risk emerges from the pricing of long-term bonds. We enrich the benchmark model with an endogenous choice of debt maturity. A calibrated fiscal rule determines government spending before bond prices are realized, so the government’s financing need is predetermined when investors bid. When the bad sunspot is realized, the long-term price collapses and the government must issue a larger quantity to clear the budget at the depressed price, retaining market access. The long-term bond therefore carries within-period confidence risk. Short-term bonds, by contrast, are not exposed to multiplicity. Their price reflects only the probability of default in the next period, so it is insulated from future borrowing.

In Cole–Kehoe rollover crises, the short-term bond is the risky one because it requires higher

payments during exclusion, and long-term debt is the insurance instrument. In our framework, the government faces a trade-off when choosing the amount of short-term bonds. Shortening maturity today reduces the government’s exposure to within-period confidence risk: less long-term debt must clear at the potentially adverse price. But more short-term debt today must be rolled over next period, raising next period’s financing need and the residual long-term issuance required tomorrow. The optimal maturity decision balances the value of insurance today against exposure tomorrow. When the probability of a confidence crisis increases, the value of this immediate insurance rises. Given the persistence of confidence risk, this can tilt the optimal portfolio toward shorter maturities, even though they entail higher future rollover costs.

We calibrate the model to features of Italy’s economy prior to the crisis and use a particle filter to infer the sequence of unobserved shocks from the data during the crisis. Our framework distinguishes two confidence shocks: the probability of a crisis and its realization. The joint dynamics of maturity shortening and spread widening identify the path of confidence risk. Our model matches spreads and maturity choice for Italy during 2010–2012 and attributes the bulk of the late-2011 spread spike to an increase in confidence risk over the episode. These results bear on the design of crisis interventions and support the policies the ECB implemented at the time.

Related Literature. Two traditions model self-fulfilling sovereign debt crises: Cole–Kehoe crises, where investors exclude the government from credit markets (Cole and Kehoe, 2000), and Calvo-style crises, where markets remain open but borrowing costs spike (Calvo, 1988; Lorenzoni and Werning, 2019). We adopt the Calvo approach, as it better captures the European experience where countries faced high yields but retained market access. Quantitative work has extended both branches: Conesa and Kehoe (2017) embed a Cole–Kehoe rollover mechanism in a model with endogenous fiscal effort and show that governments may rationally gamble for redemption, while Roch and Uhlig (2018) characterize how bailout guarantees pin down the fundamental equilibrium even when sunspots remain possible. On the Calvo side, Ayres et al. (2024) show that multiplicity is quantitatively relevant only during persistent stagnations; we complement their analysis by adding endogenous maturity choice, which provides identification of confidence shocks in the data. Aguiar et al. (2022) characterize self-fulfilling crises driven by lack of commitment at bond auctions, including “desperate deals” closely related to the bad equilibrium in our model. Auclert and Rognlie (2016) prove equilibrium is unique with one-period debt, confirming that long-term bonds are necessary for our multiplicity mechanism.

We build on the quantitative sovereign default framework: Eaton and Gersovitz (1981) and Arellano (2008) on default risk, Aguiar and Gopinath (2006) on the role of trend shocks, and Chatterjee and Eyigungor (2012) on long-term debt with default. On maturity choice, Niepelt

(2014) provides the canonical theoretical statement under lack of commitment: equilibrium maturity shortens during crises because short-term debt is less exposed to the time-inconsistency of future repayment. [Broner et al. \(2013\)](#) document the empirical counterpart in a panel of emerging economies, where issuance shifts toward short maturities when long-term risk premia spike. [Arellano and Ramanarayanan \(2012\)](#) and [Hatchondo et al. \(2016\)](#) study the insurance-dilution trade-off, and [Bigio et al. \(2023\)](#) characterize the optimal continuum-maturity issuance problem under liquidity costs. [Aguiar et al. \(2019\)](#) show the sovereign should actively manage only short-term debt under limited commitment. In our model, maturity plays a different role: it manages within-period confidence risk.

The most direct antecedent is [Bocola and Dovis \(2019\)](#), who find a modest role for confidence risk in Italian spreads using a Cole–Kehoe model with endogenous maturity; [Bocola et al. \(2019\)](#) survey the broader Eaton–Gersovitz approach to Eurozone debt and reach similar conclusions. In their Cole–Kehoe setup, higher rollover risk calls for *longer* maturity, so the same observed shortening signals low perceived risk and the quantitative weight of confidence is bounded by the rarity of observed exclusion. In our Calvo-style framework with long-term bond multiplicity, the prediction flips: higher confidence risk calls for *shorter* maturity, reconciling Italy’s observed behavior with the confidence crisis narrative. On the empirical side, [Bocola \(2016\)](#) uses a related particle-filter approach to quantify sovereign-bank feedback during the Italian crisis, while [De Grauwe and Ji \(2013\)](#) and [Krishnamurthy et al. \(2018\)](#) document the decoupling of euro-area yields from fundamentals and the large effects of ECB interventions. Within the structural European crisis literature, [Aguiar et al. \(2015\)](#) study self-fulfilling crises in monetary unions, [Bianchi and Mondragon \(2022\)](#) show lack of monetary independence amplifies rollover risk, and [Corsetti and Dedola \(2016\)](#) analyze how central bank purchases eliminate the pessimistic equilibrium. Our model abstracts from sovereign-bank feedback, a channel that further amplifies the welfare relevance of confidence risk: [Acharya et al. \(2014\)](#) document the empirical transfer of risk from banks to sovereigns, [Farhi and Tirole \(2018\)](#) provide a theoretical foundation for the doom loop, and [Brunnermeier et al. \(2016\)](#) propose European Safe Bonds as a structural response. For broader surveys, see [Aguiar and Amador \(2014\)](#) and [Aguiar et al. \(2016\)](#).

Outline. Sections 2 and 3 present the model and the government’s problem. Section 4 shows how maturity manages confidence risk, and Section 5 calibrates the model and examines its multiplicity and identification properties. Section 6 decomposes Italian spreads into fundamental and confidence components, and Section 7 concludes.

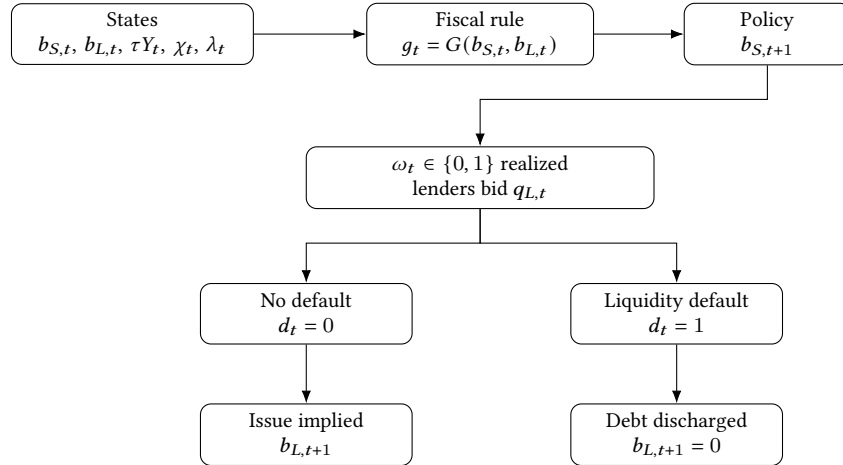
2 Model

We consider a small open economy with discrete time and an infinite horizon. The periods are indexed by $t \in \{0, 1, 2, \dots\}$. The economy is populated by a benevolent government and a continuum of risk-averse foreign investors.

2.1 Timing

We adopt the timing in [Lorenzoni and Werning \(2019\)](#). A fiscal rule pins down current expenditure before bond prices are realized, so the government cannot offset adverse price movements by cutting spending within the period. The government's only active choice is the next-period short-term debt $b_{S,t+1}$; long-term issuance is adjusted to clear the budget constraint. Within each period t , the sequence is as follows.

Figure 1: Timing



At the beginning of the period t , tax revenue τY_t (where Y_t is output and τ is a fixed tax rate), the term-premium shock χ_t , and the confidence risk shock λ_t are realized. A fiscal rule $G(\cdot)$ then establishes current spending g_t as a function of the inherited debt portfolio. Given g_t , the government chooses the next-period of short-term debt $b_{S,t+1}$. Next, the sunspot $\omega_t \in \{0, 1\}$ is realized and lenders bid for long-term bonds, determining their price $q_{L,t}$. Finally, taking the price as given, the government issues whatever long-term debt $b_{L,t+1}$ clears the budget constraint. If no feasible issuance clears the budget, the government experiences a *liquidity default* ($d_t = 1$). Long-term issuance and default are equilibrium outcomes implied by $b_{S,t+1}$ and the price of the bond, rather than independent policy choices. After default, the government enters the period $t + 1$ with zero debt.

Compared to [Lorenzoni and Werning \(2019\)](#), the government here chooses a portfolio of short- and long-term bonds. As shown by [Lorenzoni and Werning \(2019\)](#) and [Aguiar and Amador \(2020\)](#), the interaction between current prices and future borrowing in long-term bonds can generate self-fulfilling crises. [Lorenzoni and Werning \(2019\)](#) also show that with only short-term debt, the government chooses short-term issuance $b_{S,t+1}$, which has a unique price and is immune to self-fulfilling dynamics. Long-term issuance $b_{L,t+1}$ then adjusts to clear the budget constraint and remains subject to the multiplicity of the Laffer curve. This portfolio choice allows the government to manage its exposure to confidence crises.

2.2 Stochastic Processes

We model two sources of uncertainty: *fundamental risk* and *confidence risk*. Fundamental risk is summarized by the vector of economic shocks $\{y_t, \chi_t\}$, where $y_t = \log(Y_t)$ is the logarithmic output and χ_t is a term-premium shock. The two shocks are correlated and evolve according to a joint VAR(1) process:

$$\begin{bmatrix} y_{t+1} \\ \chi_{t+1} \end{bmatrix} = \begin{bmatrix} (1 - \rho_y)\mu_y \\ 0 \end{bmatrix} + \begin{bmatrix} \rho_y & 0 \\ 0 & \rho_\chi \end{bmatrix} \begin{bmatrix} y_t \\ \chi_t \end{bmatrix} + \begin{bmatrix} \epsilon_{t+1}^y \\ \epsilon_{t+1}^\chi \end{bmatrix},$$

where $[\epsilon^y, \epsilon^\chi]'$ are normally distributed with the covariance matrix Σ .

The confidence risk reflects the possibility of multiple equilibria in financial markets. The probability of a confidence crisis, λ_t , evolves as an AR(1) process:

$$\lambda_{t+1} = (1 - \rho_\lambda)\bar{\lambda} + \rho_\lambda\lambda_t + \sigma_\lambda\epsilon_{t+1}^\lambda, \quad \epsilon^\lambda \sim N(0, 1),$$

where $\bar{\lambda}$ is the long-run mean. The continuous process is discretized onto a bounded grid $[0, \lambda_{\max}]$ (see Appendix B). After the government chooses $b_{S,t+1}$ but before lenders bid for long-term bonds, a sunspot $\omega_t \in \{0, 1\}$ is realized: with probability λ_t the bad sunspot occurs ($\omega_t = 1$) and bond prices are low; with probability $1 - \lambda_t$ the good sunspot occurs ($\omega_t = 0$) and bond prices are high. Thus, λ_t captures *time-varying vulnerability* to a confidence crisis, while ω_t captures the *realized* equilibrium selection within the period.

2.3 Government

We express the model in terms of fiscal variables, as in [Bocola and Dovis \(2019\)](#), so that the primitives governing default risk, tax revenue, spending commitments, and debt service map directly to observable fiscal data.

Preferences. The government has time-separable preferences over public expenditures:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{g_t^{1-\sigma}}{1-\sigma}, \quad (1)$$

where $\beta \in (0, 1)$ is the discount factor, σ is the risk-aversion parameter, and $\mathbb{E}_t$ denotes expectations conditional on the information available at time t .

Fiscal rule. Government spending in period t follows a fiscal rule:

$$g_t = G(b_{S,t}, b_{L,t}) \equiv \bar{g} - \alpha_1 \max(b_{S,t} + b_{L,t} - \bar{b}, 0), \quad (2)$$

where $\bar{g}$ is the average expenditure, $\bar{b}$ is a reference debt level, and $\alpha_1 \geq 0$ governs fiscal tightening when total debt exceeds $\bar{b}$. The specification captures the empirical regularity that governments tighten fiscal policy when debt is elevated (Bohn, 1998).

The fiscal rule determines the government's financing need before the sunspot is drawn: because g_t is a known function of the state at the beginning of the period, the deficit $g_t - \tau Y_t$ is predetermined when lenders bid for long-term bonds. This preserves the friction of Lorenzoni and Werning (2019): the government cannot escape a bad equilibrium by committing to lower spending within the period.

Debt instruments. As in Arellano and Ramanarayanan (2012), the government issues short- and long-term bonds. The short-term bond pays $(1 + \kappa_S)$ per unit of face value at maturity, where $\kappa_S > 0$ is a fixed coupon, and its price is $q_{S,t}$. Following Hatchondo et al. (2016), long-term bonds have geometrically declining coupons: one unit issued at t implies an infinite stream of payments starting at $t + 1$,

$$\kappa_L, (1 - \delta)\kappa_L, (1 - \delta)^2\kappa_L, \dots, \quad (3)$$

where $\kappa_L > 0$ and $\delta \in (0, 1)$.¹ Long-term debt evolves according to

$$b_{L,t+1} = (1 - \delta)b_{L,t} + i_t, \quad (4)$$

¹We set κ_S and κ_L to target bond prices of 1 at par in the absence of default risk. With a stochastic discount factor, this is an approximation: coupons are calibrated using the unconditional mean discount factor as the risk-free rate, $\kappa_S = \bar{R}_f - 1$ and $\kappa_L = \delta + \bar{R}_f - 1$, where $\bar{R}_f \equiv \mathbb{E}[M_{t,t+1}]^{-1}$. This normalization is standard in the quantitative sovereign debt literature (Arellano and Ramanarayanan, 2012; Hatchondo et al., 2016).

where $b_{L,t}$ denotes the outstanding stock at t , i_t is new issuance in period t , and the long-term bond price is $q_{L,t}$.

Fiscal budget. The government enters period t with portfolio $\{b_{S,t}, b_{L,t}\}$, spending g_t determined by the fiscal rule (2), and tax revenue τY_t . Under repayment ($d_t = 0$), the period budget constraint is

$$g_t + (1 + \kappa_S)b_{S,t} + \kappa_L b_{L,t} = \tau Y_t + q_{S,t} b_{S,t+1} + q_{L,t} (b_{L,t+1} - (1 - \delta)b_{L,t}). \quad (5)$$

Spending and debt service are financed with taxes and new issuance of short- and long-term debt.

Default and recovery. We assume the government repays whenever feasible, so default occurs only when tax revenue and market financing are insufficient to satisfy the budget constraint. Let $d_t = 1$ denote default and $d_t = 0$ repayment. Under default, public spending falls to an exogenous level $g_D < \tau Y_{\min}$, where $Y_{\min}$ is the minimum realization of output. Following default, all bondholders receive a recovery value equal to a fraction ϕ of the government's net fiscal surplus, distributed pari passu across the outstanding debt stock:

$$\tilde{v}_{t+1} \equiv \frac{\phi(\tau Y_{t+1} - g_D)}{b_{S,t+1} + b_{L,t+1}}. \quad (6)$$

Both short- and long-term bondholders receive $\tilde{v}_{t+1}$ per unit of face value in default, capped at the face value of each instrument. This specification makes default costly through the reduction in public spending and the capital loss on long-term claims. The defaulting government regains access to financial markets immediately in the following period, entering with a clean balance sheet: $b_{S,t+1} = b_{L,t+1} = 0$.

2.4 International Investors

International investors are competitive and price government bonds using a stochastic discount factor (SDF) $M_{t,t+1}$. Following [Bocola and Dovis \(2019\)](#), we model changes in risk aversion as a fundamental shock that is observable and exogenous.² In particular, we model the lenders' SDF

²This specification follows [Ang and Piazzesi \(2003\)](#) and is used by [Bocola and Dovis \(2019\)](#) to allow for time-variation in risk premia on longer-maturity bonds.

as

$$M_{t,t+1} = \exp\{m_{t,t+1}\}, \quad (7)$$

$$m_{t,t+1} = -(\phi_0 + \phi_1 \chi_t) - \frac{1}{2} \kappa_t^2 + \kappa_t \varepsilon_{\chi,t+1}, \quad (8)$$

$$\chi_{t+1} = \rho_\chi \chi_t + \varepsilon_{\chi,t+1}, \quad \varepsilon_{\chi,t+1} \sim \mathcal{N}(0, 1), \quad (9)$$

$$\kappa_t = \kappa_0 + \kappa_1 \chi_t. \quad (10)$$

Short-term debt. Bond prices satisfy standard no-arbitrage conditions. Short-term bonds receive recovery $\tilde{v}_{t+1}$ per unit in default (the same rate as long-term bonds), so their price satisfies

$$q_{S,t} = \mathbb{E}_t \left[M_{t,t+1} \left((1 - d_{t+1})(1 + \kappa_S) + d_{t+1} \tilde{v}_{t+1} \right) \right]. \quad (11)$$

Long-term debt. Conditional on repayment, one unit pays the coupon κ_L next period and leaves a fraction $(1 - \delta)$ of the claim outstanding, which trades at $q_{L,t+1}$. Under default, the bond receives $\tilde{v}_{t+1}$ per unit of face value. Therefore,

$$q_{L,t} = \mathbb{E}_t \left[M_{t,t+1} \left((1 - d_{t+1})(\kappa_L + (1 - \delta)q_{L,t+1}) + d_{t+1} \tilde{v}_{t+1} \right) \right]. \quad (12)$$

Expectations in (11)–(12) integrate over both fundamental shocks and equilibrium selection in financial markets, so confidence risk affects bond prices through the distribution of future default and continuation values.

2.5 Competitive Equilibrium

Definition 1 (Competitive Equilibrium). Given initial government debt $\{b_{S,0}, b_{L,0}\}$, the fiscal rule $G(\cdot)$, the government's short-term debt policy $\{b_{S,t+1}\}_{t=0}^\infty$, and the exogenous processes $\{Y_t, \chi_t, \lambda_t, \omega_t\}_{t=0}^\infty$, a *competitive equilibrium* is a sequence of bond prices, long-term issuance, and default outcomes $\{q_{S,t}, q_{L,t}, b_{L,t+1}, d_t\}_{t=0}^\infty$ such that:

- i *Default.* Default in period t occurs if and only if no feasible $b_{L,t+1} \geq 0$ satisfies the budget constraint (5) at the prevailing price $q_{L,t}$: $d_t = 1\{\text{no feasible } b_{L,t+1} \text{ exists}\}$. When multiple feasible $b_{L,t+1}$ exist, the sunspot $\omega_t \in \{0, 1\}$ selects between them.
- ii *Bond pricing.* Conditional on the equilibrium default and issuance outcomes, bond prices satisfy (11) and (12), where recovery $\tilde{v}_{t+1}$ is determined by (6) given the equilibrium debt portfolio $(b_{S,t+1}, b_{L,t+1})$.

- iii *Market clearing.* In each period, bond markets clear: the quantities of short- and long-term claims held by investors equal the quantities issued by the government, $\{b_{S,t+1}, b_{L,t+1}\}$.

3 Optimal Policy

We focus on a Markov equilibrium with payoff-relevant state (s, λ) , where $s \equiv (b_S, b_L, y, \chi)$ collects the inherited debt portfolio (b_S, b_L) , the output shock y , and the term-premium shock χ , and λ is the time-varying probability of the bad equilibrium. The government's only choice variable is next-period short-term debt b'_S . The fiscal rule $G(b_S, b_L)$ pins spending g . Long-term issuance b'_L and default d are equilibrium outcomes: given b'_S , default occurs if no feasible b'_L clears the budget at the prevailing price, and conditional on repayment, b'_L is the issuance that clears the budget.

3.1 The Laffer Curve

We conjecture that bond price schedules $Q_S(y, \chi, \lambda, b'_S, b'_L)$ and $Q_L(y, \chi, \lambda, b'_S, b'_L)$ exist. Bond prices depend on the current fundamental shocks (y, χ) , confidence risk λ , and new issuance (b'_S, b'_L) . Because λ follows a persistent AR(1) process, the current λ affects the distribution of next-period λ' and therefore enters bond prices. Given these price schedules, define the government's net financial revenue from issuing (b'_S, b'_L) as

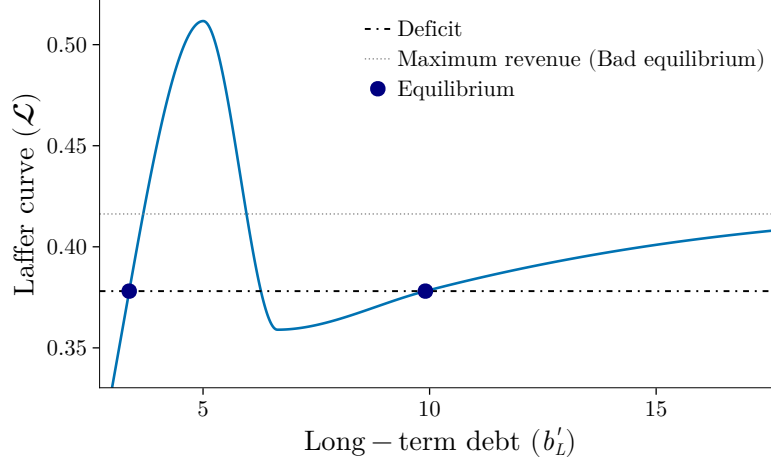
$$\mathcal{L}(s, \lambda, b'_S, b'_L) = Q_L(y, \chi, \lambda, b'_S, b'_L)(b'_L - (1 - \delta)b_L) + Q_S(y, \chi, \lambda, b'_S, b'_L)b'_S - (1 + \kappa_S)b_S - \kappa_L b_L. \quad (13)$$

Figure 2 illustrates a typical shape for the Laffer curve, $\mathcal{L}(b'_L; s, \lambda, b'_S)$, defined in equation (13). The curve captures the net revenue the government can raise from the bond market as a function of its long-term issuance b'_L , given the state (s, λ) and its short-term debt choice b'_S .

The non-monotonic shape of the Laffer curve reflects the tension between two forces. On the one hand, issuing more long-term debt mechanically raises revenue: the *quantity effect*. On the other hand, higher issuance worsens rollover risk, depressing bond prices and reducing the revenue per unit issued: the *price effect*. On the upward-sloping portion of the curve, the quantity effect dominates: markets are unconcerned about default, prices remain high, and more debt means more revenue. Beyond a threshold, the price effect takes over: bond prices collapse faster than the debt stock grows, and total revenue falls even as the face value of debt increases. As

in Lorenzoni and Werning (2019), the interaction between these two forces generates the non-monotonic shape shown in Figure 2.

Figure 2: Laffer curve at a state with two stable equilibria



Notes: The Laffer curve $\mathcal{L}(b'_L; s, \lambda, b'_S)$ as a function of long-term issuance b'_L . Dash-dotted line: financing need $g - \tau Y$. Dotted line: maximum revenue $\mathcal{L}_\infty$, the asymptote of the recovery branch (see equation (14)). Open circle: good equilibrium ($\omega = 0$); filled circle: bad equilibrium ($\omega = 1$).

The second upward-sloping portion of the Laffer curve, at high b'_L , reflects the dilution of existing creditors' recovery claims. The recovery pool $\phi(\tau Y' - g_D)$ is shared pari passu across all outstanding debt, so per-unit recovery is $\phi(\tau Y' - g_D)/(b'_S + b'_L)$. As new issuance grows, current-period issuance $b'_L - (1 - \delta)b_L$ captures a larger share of this fixed pool, generating a second increasing region in $\mathcal{L}$ even though the total pool is unchanged. It is this second branch that sustains the bad (high-issuance) equilibrium in the Lorenzoni–Werning sense.

This structure allows for multiple equilibria. An equilibrium requires the government's net market revenue to equal its financing need, $g - \tau Y$. Graphically, an equilibrium is an intersection between the Laffer curve and the horizontal line at height $g - \tau Y$. For a range of deficits, multiple stable intersections coexist.

Price of short-term bonds Because short-term bonds are traded before the sunspot ω is realized, their price must already reflect the possibility of either equilibrium. Thus, for any candidate short-term issuance b'_S , lenders set the price as the λ -weighted average of the short-term bond prices in the good and bad equilibria:

$$Q_S(y, \chi, \lambda, b'_S) = (1 - \lambda) Q_S(y, \chi, \lambda, b'_S, b_{L,g}^*) + \lambda Q_S(y, \chi, \lambda, b'_S, b_{L,b}^*),$$

where $b_{L,g}^*$ and $b_{L,b}^*$ are the equilibrium levels of long-term debt in the good and bad equilibria.

Laffer asymptote As b'_L goes to ∞ , default becomes certain and the per-unit bond price approaches the discounted recovery value: Q_L approaches to $\mathbb{E}[M(\chi') \phi(\tau Y' - g_D)/b'_L]$. Total long-term revenue $Q_L(b'_L - (1-\delta)b_L)$ therefore converges to the discounted recovery pool $\mathbb{E}[M(\chi') \phi(\tau Y' - g_D)]$. Define the *Laffer asymptote*

$$\mathcal{L}_\infty(s, \lambda, b'_S) \equiv \mathbb{E}[M(\chi') \phi(\tau Y' - g_D)] + Q_S(y, \chi, \lambda, b'_S) b'_S - (1 + \kappa_S) b_S - \kappa_L b_L, \quad (14)$$

which is the maximum net revenue the government can raise from the bond market in the limit of unlimited long-term issuance. When $\mathcal{L}_\infty < g - \tau Y$, no high-issuance level of long-term borrowing can clear the budget if the bad sunspot ($\omega = 1$) is realized.

3.2 Default and Issuance Rules

An equilibrium requires a value of b'_L at which $\mathcal{L}(s, \lambda, b'_S, b'_L) = g - \tau Y$. Following [Lorenzoni and Werning \(2019\)](#), only intersections on locally *increasing* portions of $\mathcal{L}$ are stable equilibria; intersections on decreasing portions are unstable and discarded. Let

$$\mathbb{B}_L^+(s, \lambda, b'_S) = \left\{ b'_L : \mathcal{L}(s, \lambda, b'_S, b'_L) = g - \tau Y, \quad \frac{\partial \mathcal{L}}{\partial b'_L} > 0 \right\} \quad (15)$$

denote the set of stable budget-clearing levels.

Given $\mathbb{B}_L^+(s, \lambda, b'_S)$, the sunspot ω selects which price prevails. The default indicator is

$$D(s, \lambda, \omega, b'_S) = \begin{cases} 1, & \mathbb{B}_L^+(s, \lambda, b'_S) = \emptyset, \\ 1, & \omega = 1 \text{ and } \mathcal{L}_\infty(s, \lambda, b'_S) < g - \tau Y, \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

The rule has two parts. A *liquidity default* arises when no price clears the budget, $\mathbb{B}_L^+(s, \lambda, b'_S) = \emptyset$: the government cannot finance the deficit at any price and defaults under either sunspot. A *Cole–Kehoe multiplicity* arises when the deficit is financeable at a high price, but not a low one. The good sunspot ($\omega = 0$) selects the high price and the government repays; the bad sunspot ($\omega = 1$) selects a low price at which no issuance clears, $\mathcal{L}_\infty(s, \lambda, b'_S) < g - \tau Y$, and the government defaults on that period. Also, the long-term equilibrium debt rule is

$$B_L(s, \lambda, \omega, b'_S) = \begin{cases} \max \mathbb{B}_L^+(s, \lambda, b'_S), & \text{if } \omega = 1 \text{ and } D(s, \lambda, 1, b'_S) = 0, \\ \min \mathbb{B}_L^+(s, \lambda, b'_S), & \text{if } \omega = 0 \text{ and } D(s, \lambda, 0, b'_S) = 0, \\ 0, & \text{if } D(s, \lambda, \omega, b'_S) = 1. \end{cases} \quad (17)$$

Calvo-type of multiplicity arises when $\mathbb{B}_L^+(s, \lambda, b'_S)$ contains more than one element: multiple stable long-term issuance levels clear the budget, and the sunspot selects between them. By contrast, short-term debt is priced by international investors and, given government policies and the default rule, its price is uniquely pinned down. This asymmetry makes long-term issuance the margin that generates confidence-driven fluctuations in prices, while the government can use short-term issuance to reduce confidence risk.

3.3 Value Function

Given the rules for default and long-term debt, the value function of the government solves

$$V(s, \lambda) = \max_{b'_S} \mathbb{E}_{y', \chi', \lambda', \omega} \left[(1 - d)(u(g) + \beta V(b'_S, b'_L, y', \chi', \lambda')) + d(u(g_D) + \beta V(0, 0, y', \chi', \lambda')) \right], \quad (18)$$

$$\text{s.t.} \quad d = D(s, \lambda, \omega, b'_S), \quad b'_L = B_L(s, \lambda, \omega, b'_S).$$

Equation (18) states that the government chooses b'_S taking prices as given; default d and long-term debt b'_L are implied by budget feasibility and the equilibrium long-term bond price. The period payoff is $u(g)$ under repayment and $u(g_D)$ under default, and the next-period state resets to zero debt after default, consistent with the timing of Sections 2 and 3. The solution yields a decision rule for short-term debt,

$$b'_S = \mathcal{B}_S(s, \lambda).$$

3.4 Price Functions

In a Markov equilibrium, the conjectured bond price schedules must satisfy competitive investors' no-arbitrage conditions, given the equilibrium law of motion implied by government policies and

the default and issuance rules. Using the SDF $M(\chi)$, bond prices satisfy:

$$Q_S(y, \chi, \lambda, b'_S, b'_L) = \mathbb{E} \left[M(\chi, \chi') \left((1 - d') (1 + \kappa_S) + d' \tilde{v}' \right) \right], \quad (19)$$

$$Q_L(y, \chi, \lambda, b'_S, b'_L) = \mathbb{E} \left[M(\chi, \chi') \left\{ (1 - d') (\kappa_L + (1 - \delta) Q_L(y', \chi', \lambda', b''_S, b''_L)) + d' \tilde{v}' \right\} \right], \quad (20)$$

where $d' = D(s', \lambda', \omega', b''_S)$, $b''_S = \mathcal{B}_S(s', \lambda')$, $b''_L = B_L(s', \lambda', \omega', b''_S)$, and $s' = (b'_S, b'_L, y', \chi')$. The price of the continuation bond $Q_L(y', \chi', \lambda', b''_S, b''_L)$ is evaluated in the state of the next-period (y', χ', λ') and the portfolio (b''_S, b''_L) chosen in s' . Both Q_S and Q_L are conditional on no current-period default: under $d = 1$ the government enters autarky with zero issuance.

3.5 Markov Equilibrium

Definition 2 (Markov Perfect Equilibrium). Given an initial state s_0 , the laws of motion for (y, χ, λ) , the sunspot distribution $\omega \mid \lambda$, and the fiscal rule $G(\cdot)$, a Markov perfect equilibrium consists of functions

$$V, \mathcal{B}_S, D, B_L, Q_S, Q_L$$

such that:

- i) **Feasibility and selection:** Default is given by (16), and conditional on repayment, long-term debt satisfies (17).
- ii) **Government optimality:** Given (Q_S, Q_L) and the induced mappings D and B_L , V solves the Bellman equation (18) and $\mathcal{B}_S$ is the associated optimal policy rule.
- iii) **Pricing:** Given $(\mathcal{B}_S, D, B_L)$, bond prices satisfy the no-arbitrage conditions (19)–(20).

In equilibrium, government portfolio choices are optimal given prices, prices are consistent with competitive pricing given policies, and the sunspot selects among long-term issuance levels that clear the fiscal budget when multiplicity arises.

3.6 Multiplicity

We close the section by characterizing the multiplicity region as a function of the state and the government's portfolio choice. Let $\mathbb{B}_L^+(s, \lambda, b'_S)$ denote the set of stable budget-clearing long-term issuance levels defined in (15). The state-policy space partitions into three regions, distinguished by how many stable equilibria exist and whether the bad sunspot triggers default. In the **safe region** $\mathcal{M}_S$, one stable equilibrium sits on the low-issuance branch and the bad sunspot does

not default; equilibrium features low long-term debt and high bond prices, with ω irrelevant. In the **Multiplicity - Run region** $\mathcal{M}_R$, one stable good equilibrium exists but the Laffer curve lies below the deficit at all high issuance levels, so the bad sunspot cannot clear the budget and defaults immediately, in the spirit of [Cole and Kehoe \(2000\)](#). In the **Multiplicity - Low price region** $\mathcal{M}_M$, two stable equilibria coexist: one with low issuance and high prices, the other with high issuance and low prices, with the sunspot ω selecting between them via the [Lorenzoni and Werning \(2019\)](#) mechanism. Formally:

Definition 3 (Equilibrium regions).

$$\begin{aligned}\mathcal{M}_S &\equiv \{(s, \lambda, b'_S) : |\mathbb{B}_L^+(s, \lambda, b'_S)| = 1, \quad D(s, \lambda, 1, b'_S) = 0\}, \\ \mathcal{M}_R &\equiv \{(s, \lambda, b'_S) : \mathbb{B}_L^+(s, \lambda, b'_S) \neq \emptyset, \quad D(s, \lambda, 1, b'_S) = 1\}, \\ \mathcal{M}_{LP} &\equiv \{(s, \lambda, b'_S) : |\mathbb{B}_L^+(s, \lambda, b'_S)| = 2\}.\end{aligned}$$

Since bond prices depend on the current λ (which forecasts future crisis risk), the boundaries of all three regions shift with λ . Output operates analogously: higher y simultaneously reduces the financing need $g - \tau Y$ and increases the prices of bonds through improved expected repayment, two strengthening forces that push the economy towards the safe region $\mathcal{M}_S$.

4 Short-term debt and confidence risk

This section studies how short-term debt shifts the boundaries of this set. The government shifts the boundaries of the multiplicity region by changing its short-term issuance b'_S . Financing a larger share of the deficit with short-term bonds reduces the long-term issuance the government must place at settlement. Less long-term issuance clears the budget even when long-term prices fall, so the bad equilibrium becomes harder to sustain. A high enough b'_S removes the second budget-clearing root and moves the state into the safe region. Short-term debt insures the government against the within-period confidence crisis.

We formalize this mechanism in a stylized infinite-horizon environment. The stylized environment isolates the insurance channel and allows an analytical characterization of the conditions under which short-term debt mitigates confidence risk.

A stylized environment. Consider an infinite-horizon version of the model in which the output and the term-premium shock are constant, $Y_t = \bar{Y}$ and $\chi_t = \bar{\chi}$, and the confidence-risk probability is a constant parameter $\lambda \in (0, 1)$. The fiscal rule, the default rule, and the recovery

rule are unchanged. We further assume that the period utility is linear $u(g) = g$, so that the marginal value of consumption is constant: the long-bond stock does not raise the marginal cost of carrying short-term debt, and the within-period insurance channel of b'_S operates without a competing precautionary motive. We restrict attention to a stationary equilibrium in which the government chooses a constant short-term issuance b_S^* each period, the long-debt level alternates between two values b_L^g and b_L^b according to the realized sunspot, and the bond prices at the two equilibrium roots, Q_S , Q_L^g and Q_L^b , are constants determined by stationary fixed-point conditions. The bad-sunspot root sits on the recovery branch of the Laffer curve; the good-sunspot root is repaid this period, and the bond's one-period-ahead payoff is a λ -weighted average of its good-state and bad-state values:

$$Q_L^b = \tilde{v} \equiv \frac{\phi(\tau\bar{Y} - g_D)}{b_S^* + b_L^b}, \quad (21)$$

$$Q_L^g = \beta[(1 - \lambda)(\kappa_L + (1 - \delta)Q_L^g) + \lambda \tilde{v}]. \quad (22)$$

The price of the short-bond Q_S is quoted at the auction, before ω is drawn. The bond matures one period later, paying $1 + \kappa_S$ if the government repays and the per-unit recovery $\tilde{v}$ if it defaults. With $\omega = 1$ triggering the bad-equilibrium outcome on the recovery floor, the short-bond price satisfies

$$Q_S = \beta[(1 - \lambda)(1 + \kappa_S) + \lambda \tilde{v}]. \quad (23)$$

Proposition 1 (Insurance value of short-term debt, stylized environment). *In the stylized stationary equilibrium, suppose that the primitives satisfy*

$$\beta\kappa_L > \tilde{v}[1 - \beta(1 - \delta)],$$

Then the optimal short-term issuance is strictly increasing in confidence risk,

$$\frac{\partial b_S^*}{\partial \lambda} > 0.$$

Proof. See Appendix A. □

In this environment, the long bond is worth more in the good sunspot than in the bad one: its good-state price is Q_L^g , while in the bad state it only trades at its recovery value $Q_L^b = \tilde{v}$. The reason is that, in a good sunspot, the bond is fully serviced: it pays a coupon κ_L every period, and a fraction $1 - \delta$ of the principal survives and can be sold again at the same price next period.

Intuitively, the investor is holding a perpetuity that keeps generating payments and retains resale value, so the bond is valuable. In contrast, in a bad sunspot the bond is marked down to its recovery value $\tilde{v}$: investors expect to get only this reduced amount back, so the bond is strictly less valuable.

As the perceived risk λ of the bad sunspot rises, investors attach more weight to this low recovery value and less to full repayment. This makes the good-state price $Q_L^g(\lambda)$ drift downward toward $\tilde{v}$: even in the good sunspot, people price the bond more cautiously because they increasingly anticipate the possibility of ending up in the bad sunspot.

This gap in values gives short-term debt an insurance role. From the government's budget identity, issuing one more unit of short-term debt allows the government to buy back Q_S/Q_L^ω units of long-term debt in sunspot state ω . Because the long bond is cheaper in the bad sunspot ($Q_L^b = \tilde{v} < Q_L^g$), each unit of short-term debt lets the government issue less long-term debt precisely in the bad state, when the long-bond market is weakest and long-term borrowing is most costly.

Full model. This insurance comes at a cost. The short-term bond matures in the next period and is paid in full $(1 + \kappa_S) b'_S$, so a larger short position increases the need for financing in the next period. A larger financing need raises the long-term issuance the government must place next period, and long-term issuance is the margin that carries confidence risk. So in the general model, more short-term debt today does not remove the government's exposure to a confidence crisis; it moves that exposure from the current period to the next. The maturity choice trades insurance today against exposure tomorrow.

When the probability of a confidence crisis λ increases, the value of insuring the current period increases relative to the discounted future, and the government tilts the portfolio towards short-term debt. Persistence in λ tempers this response. A high λ today forecasts a high λ tomorrow, so the shortening that ensures that the current period sets up a larger rollover when the next period is also vulnerable. Short-term issuance increases with λ without jumping to a corner, and our quantitative analysis traces this balance through Italy's 2010–2012 episode. In [Appendix A](#) we study the full model and establish some numerical conditions for such that higher short term bonds reduce the exposure of the government to multiple equilibria.

5 Quantitative Analysis

We calibrate the model to match long-run averages of the Italian economy, taking the SDF, term-premium-process, and discount-factor parameters directly from [Bocola and Dovis \(2019\)](#) for comparability. We first describe the calibration, then verify that the calibrated model exhibits

Laffer-curve multiplicity at Italian debt levels, and document the government’s optimal maturity response to rising crisis probability. Finally, we use the calibrated model to decompose Italian sovereign spreads during the 2010–2012 crisis.

5.1 Calibration

Our calibration targets three long-run Italian moments: the average debt-to-GDP ratio, the average sovereign spread, and the average debt maturity. The fiscal-rule parameters $(\bar{g}, \alpha_1, \bar{b})$, the recovery rate ϕ , the default-period spending g_D , and the confidence-risk process parameters $(\lambda_{\max}, \rho_\lambda)$ are jointly calibrated by minimizing the GMM distance between the simulated and empirical moments, subject to the structural side-restriction that the calibrated state admits the Laffer-curve multiplicity at Italian debt levels (discussed below). We fix the remaining parameters at conventional values or directly empirical counterparts.

The baseline parameters are in Table 1. The utility curvature parameter is $\sigma = 2$, a standard value in macroeconomic models. The long-term bond decay is $\delta = 0.033$, implying an average duration of 7.6 years. The tax rate is $\tau = 0.41$, matching the average tax revenue-to-GDP ratio in Italy.

The fiscal rule parameters $(\bar{g}, \alpha_1, \bar{b})$ require special treatment because they serve two functions simultaneously. First, they must generate plausible unconditional fiscal dynamics: $\bar{g}$ is the long-term spending level the rule prescribes at the reference debt level $\bar{b}$, and α_1 governs the speed of fiscal consolidation in response to debt deviations. Second, α_1 must be small enough that the calibrated state lies in $\mathcal{S}_M$ or $\mathcal{S}_{CK}$ (defined in Section 4) at debt levels representative of Italy during 2008–2012. This second requirement is structural rather than statistical: if α_1 is too large, the fiscal rule commits to such deep austerity that the deficit can always be financed at the good equilibrium regardless of the sunspot realization, eliminating the multiplicity that drives our results. We therefore choose $(\bar{g}, \alpha_1, \bar{b})$ jointly to satisfy both requirements and verify that the calibrated values place the modal Italian state in $\mathcal{S}_M$ or $\mathcal{S}_{CK}$ (the filter that recovers this state is introduced in Section 6).

We estimate the GDP-process parameters from detrended Italian GDP data. We take the parameters of SDF, term-premium-process, and discount-factor from [Bocola and Dovis \(2019\)](#). In particular, we set $\beta = 0.98$ following [Bocola and Dovis 2019](#) and verify that model moments are insensitive to this value: because default in our model is a liquidity event determined by clearing the bond market rather than a strategic choice, β does not govern the default threshold as does in strategic default models.

Table 1: Model Parameters

Parameter	Symbol	Value	Target / Source
<i>Predetermined parameters</i>			
Risk aversion	σ	2	Conventional value
Long-term bond decay	δ	0.033	Long-term bond duration (≈ 7.6 years)
Tax rate	τ	0.41	Italian tax revenues / GDP
Discount factor	β	0.98	Bocola and Dovis (2019)
<i>GDP process</i>			
GDP persistence	ρ_y	0.9668	Italian GDP (detrended)
GDP mean	μ_y	-0.0002	Italian GDP (detrended)
GDP std. dev.	σ_y	0.008	Italian GDP (detrended)
<i>SDF and term-premium process (taken from Bocola and Dovis (2019))</i>			
SDF intercept	ϕ_0	0.005	Bocola and Dovis (2019)
SDF slope	ϕ_1	0.002	Bocola and Dovis (2019)
Price-of-risk intercept	κ_0	0.161	Bocola and Dovis (2019)
Price-of-risk slope	κ_1	0.374	Bocola and Dovis (2019)
Term-premium persistence	ρ_χ	0.513	Bocola and Dovis (2019)
Term-premium std. dev.	σ_χ	1.0	Bocola and Dovis (2019)
<i>Calibrated via method of simulated moments</i>			
Recovery rate	ϕ	0.50	Method of simulated moments
Default spending	g_D	0.013	Method of simulated moments
Average spending	$\bar{g}$	0.46	Method of simulated moments
Fiscal austerity slope	α_1	0.09	Method of simulated moments
Reference debt level	$\bar{b}$	2.8	Method of simulated moments
Crisis prob. upper bound	$\lambda_{\max}$	0.08	Method of simulated moments
Crisis prob. persistence	ρ_λ	0.5	Method of simulated moments

We calibrate the fiscal-rule parameters $(\bar{g}, \alpha_1, \bar{b})$ as described above, not from a regression but to satisfy both the moment targets and the multiplicity requirement at Italian debt levels. The recovery rate ϕ , the spending in the default-period g_D , and the confidence-risk limits $(\lambda_{\min}, \lambda_{\max}, \rho_\lambda)$ are calibrated to meet three empirical targets: the average debt-to-GDP ratio, the average sovereign spread and the average debt maturity. Appendix C provides empirical definitions of these variables.

Table 2: Empirical Targets: Data vs. Model

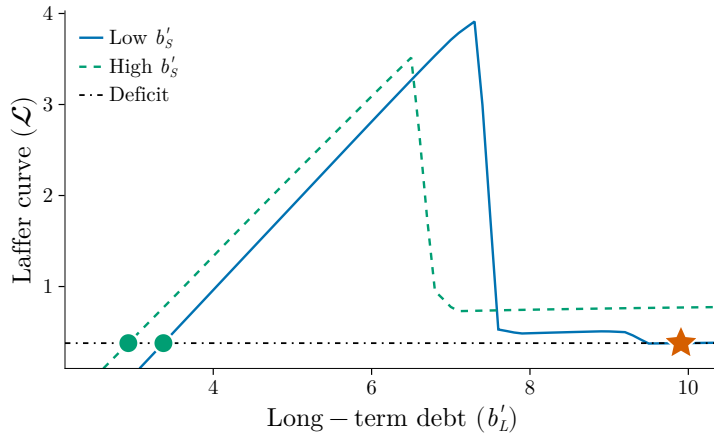
Moment	Data	Model
Average debt-to-GDP ratio (%)	87.9	82.1
Average spread (basis points)	61.0	60.7
Average debt maturity (years)	6.8	7.1

The model matches the average spread almost exactly (60.7 vs. 61 bp), produces a debt-to-GDP ratio of 82.1 percent against 87.9 in the data, and an average maturity of 7.1 years against 6.8 in the data.

5.2 Short-Term Debt and Multiplicity

At Italy's median debt, increasing short-term issuance from low to high levels eliminates the bad-equilibrium intersection of the Laffer curve (Figure 3). With low short-term debt (solid blue), the deficit line intersects the Laffer curve at two stable points: the good equilibrium (low issuance, high price, open circle) and the bad equilibrium (high issuance, low price, filled circle). With high short-term debt (dashed green), only one stable intersection survives. By financing a larger share of the deficit through short-term bonds, whose price is predetermined before the sunspot ω is drawn, the government reduces the long-term issuance required to clear the budget at settlement, making the bad equilibrium infeasible. This is the quantitative counterpart of the mechanism formalized in Proposition 1.

Figure 3: Short-term debt and equilibrium multiplicity

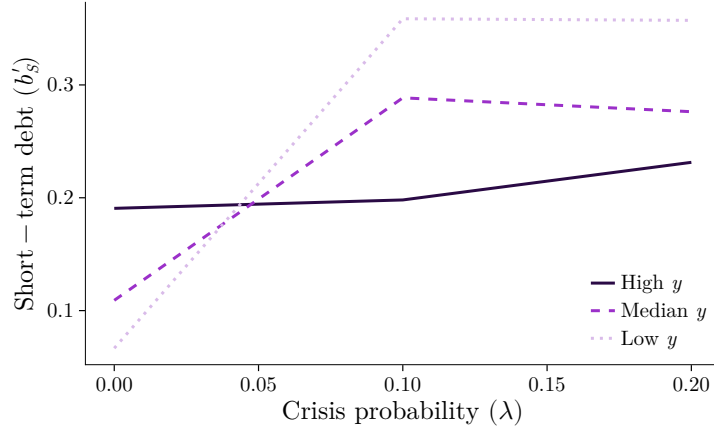


Notes: The Laffer curve $\mathcal{L}(b'_L; s, \lambda, b'_S)$ is plotted at Italy's median estimated debt portfolio (recovered by the particle filter) for two levels of short-term debt b'_S . Stable budget-clearing intersections are marked with an open circle (good equilibrium) and a filled circle (bad equilibrium) for the low b'_S curve (blue, solid). The high b'_S curve (orange, dashed) has a unique equilibrium marked with a cross.

5.3 Optimal Maturity Response

As confidence risk λ rises, the government shortens maturity to insure against the bad equilibrium (Figure 4). Once b'_S is large enough to eliminate that equilibrium, the insurance motive disappears and issuance flattens or declines.

Figure 4: Short-term debt policy: response to rising crisis probability



Notes: Expected short-term debt $E[b'_S | s, \lambda]$ as a function of crisis probability λ , for three output levels (high, median, low y) at the representative calibrated debt state.

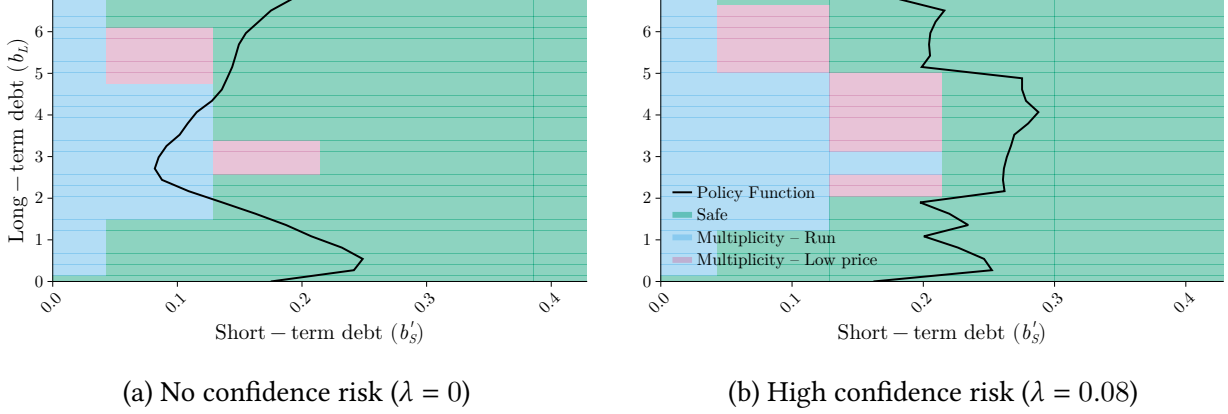
The maturity choice balances two forces that pull in opposite directions across output. The first is the cost of rolling over short-term debt, which the government repays in full the next period. Rolling it over is expensive when output is low, because output is persistent: a weak year today signals a high financing need and low bond prices next year. The government therefore borrows short when output is high and long when output is low. The second force is insurance against the bad equilibrium. Weak output places the government close to the crisis region, so shortening maturity is most valuable when output is low. At low λ the rollover force dominates and short-term debt rises with output. As λ rises, insurance takes over and the ranking flips: the low-output government issues the most short-term debt. Because the two forces pull in opposite directions, the lines cross.

The insurance motive is strongest at intermediate and high b_L , where the crisis regions are wider (Figure 5). We map these regions in the space of short- and long-term debt (b'_S, b_L), in the representative output and debt state, for low and high probability of crisis λ . The background color marks the case the government faces in each (b'_S, b_L): green is the safe region, blue is the run region (Cole–Kehoe), and pink is the low-price region (Lorenzoni–Werning). The black curve is the expected short-term issuance of the government $E[b'_S | b_L, \lambda]$ at each level of long-term debt.

Two results stand out. First, the safe region (green) lies to the right: a high enough b'_S moves the economy out of both crisis regions, whatever the level of long-term debt b_L . This is the calibrated counterpart of Proposition 1: short-term debt lowers the risk of a self-fulfilling crisis because it leaves less long-term debt to be sold at adverse prices. Second, the policy curve changes to the right as λ increases: the government holds more short-term debt to reach the safe region.

The shift is largest at intermediate and high b_L , where the crisis regions are the largest and the insurance value of the shortening is greatest.

Figure 5: Multiplicity landscape and optimal short-term debt policy

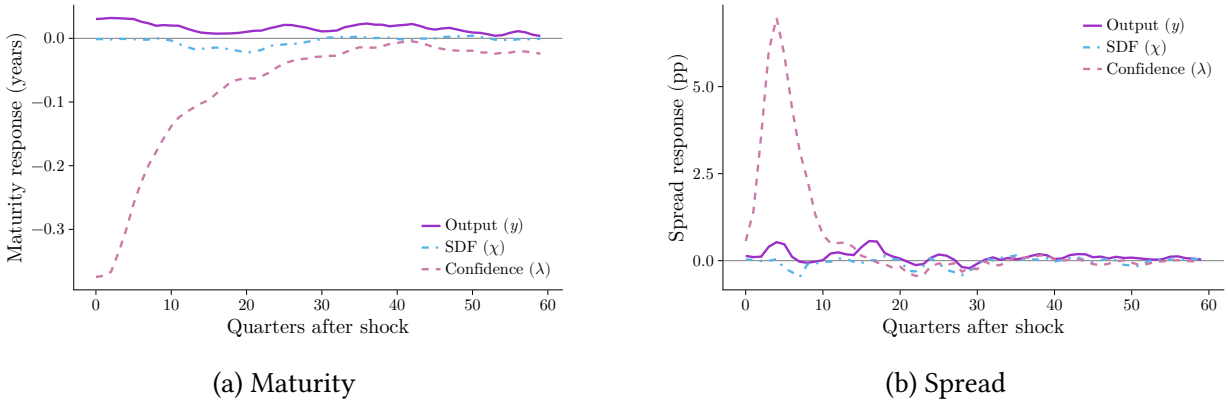


Notes: Background colors indicate the Laffer-curve case at each state-policy pair (b'_S, b_L) : green = Safe; blue = Multiplicity – Run (Cole–Kehoe); pink = Multiplicity – Low price (Lorenzoni–Werning). The black curve is $E[b'_S | b_L, \lambda]$, the expected short-term debt the government chooses given the inherited long-term debt state b_L . All other state variables held at Italy's median estimated debt portfolio.

5.4 Identification

In the model, short term bonds responds differently to a confidence shock than to a fundamental shock, and this difference identifies the confidence risk in the data. Figure 6 reports generalized impulse responses of the ergodic state to three adverse shocks: a rise in the probability of a confidence crisis λ , a decrease in output y , and a rise in the term premium χ .

Figure 6: Impulse Responses to Confidence and Fundamental Shocks



Notes: Generalized impulse responses from the ergodic state to three adverse shocks: a rise in the confidence-crisis probability λ to its grid maximum (dashed pink), a two-standard-deviation fall in output y (solid purple), and a two-standard-deviation rise in the term premium χ (dash-dot blue). The horizon is 60 quarters.

A confidence shock raises the probability of the bad equilibrium, and the government substitutes toward short-term debt to insure against it. Maturity falls by 0.38 years on impact (Figure 6, Panel a). A drop in output (a recession) raises the government’s cost of rolling over short-term debt. Because recessions persist, the government will need more funds next period while investors are less willing to pay high bond prices, making short-term refinancing costlier. As a response, the government issues more long-term debt, increasing the average maturity of its debt by about 0.03 years. A term-premium shock affects maturity if it moves the price of one bond more than the other. Such a disproportionate effect would fall on long-term bonds, which carry the term premium, and would tilt the government toward short-term debt. Numerically this effect is small, and the effect on maturity is statistically zero in all periods.

The confidence response is more than ten times larger than either fundamental response and carries the opposite sign from the output response. The spread rises by seven percentage points after the confidence shock (Panel b), by half a percentage point after the output shock, and by zero after the term-premium shock. Maturity separates confidence risk from fundamentals: a shortening of maturity alongside a widening spread identifies a rise in λ , while a widening spread with unchanged maturity points to fundamentals. The sign of the maturity response reverses the identification in [Bocola and Dovis \(2019\)](#), where higher rollover risk calls for longer maturity. Appendix E reports the debt-to-GDP responses.

6 Decomposing Italian Spreads

We now apply the model to interpret the 2008–2012 Italian sovereign debt crisis. Following [Bocola and Dovis \(2019\)](#), we cast the model as a non-linear state-space system and use a Sequential Monte Carlo particle filter to estimate the unobserved states. The fundamental side of our filter matches theirs: output and the term premium directly identify fundamental risk. The difference lies in confidence: our Calvo-style mechanism allows crises without default. Our timing also makes maturity shortening a signal of increased risk of confidence, reversing the identification in [Bocola and Dovis \(2019\)](#).

6.1 The State-Space Representation

The observation and transition equations are

$$\begin{aligned} X_t &= h(S_t) + \eta_t, \\ S_t &= f(S_{t-1}, \epsilon_t), \end{aligned}$$

where the functions h and f are defined by the model's equilibrium conditions and policy functions, and η_t is a vector of measurement errors.

The vector of observables X_t contains four variables: detrended GDP (y_t), the term premium factor (χ_t), debt maturity, and the sovereign spread. The state vector S_t collects all variables necessary to describe the economy: the two observable fundamentals (y_t, χ_t), the two unobserved confidence shocks (λ_t, ω_t), and the endogenous variables determined by government policy in the previous period ($b_{S,t}, b_{L,t}$).

Following [Bocola and Dovis \(2019\)](#), we set the measurement error variance for y_t and χ_t to zero, so that the filtered path of fundamentals coincides exactly with the data. This leaves two observables (maturity and spreads) and two unobserved shocks (λ_t and ω_t) for the filter to recover. We allow for small measurement errors in maturity and spreads; [Appendix D](#) details the algorithm and parameter values.

The filtering procedure delivers, for each period, estimated distributions of the unobserved states. Two objects are of particular interest: the probability of a confidence crisis (λ_t), which governs the government's exposure to a confidence crisis, and the sunspot realization itself (ω_t), which determines whether the crisis materializes. The model's timing provides an identification scheme that lets the filter separate the two shocks using the observed dynamics of maturity and spreads.

6.2 Spread Decomposition

The main goal is to decompose the observed Italian sovereign spread into components driven by economic fundamentals ($\lambda_t = 0, \omega_t = 0$) and by self-fulfilling crises. The *confidence crisis component* is the difference between the spread implemented in the full model and the counterfactual $\omega_t = 0$ and $\lambda = 0$. It captures the effect of confidence risk on the spread of government bonds.

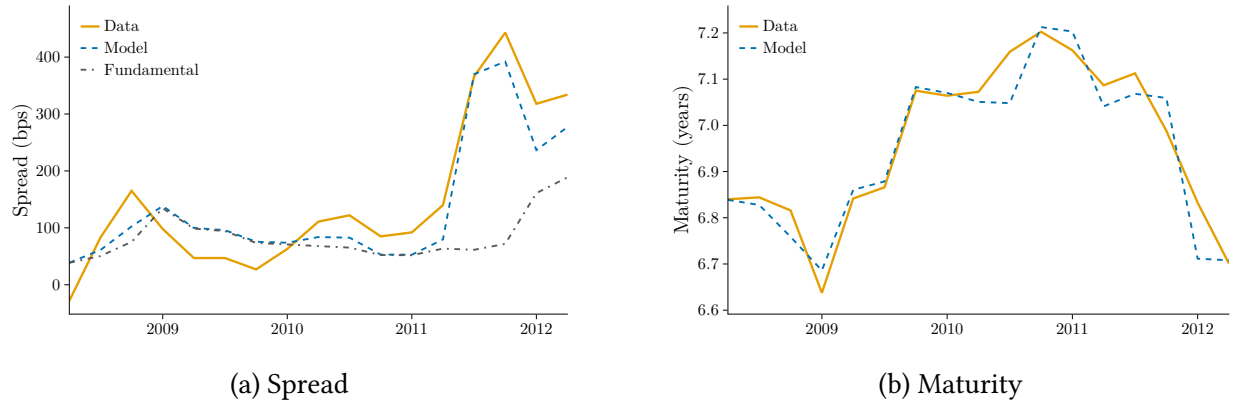
Fundamental dynamics. The filtered fundamental shocks deteriorate during the episode ([Figure 7](#)). Italian output falls from 0.04 percent above trend at the start of the sample to -0.04 percent below trend by 2012, reflecting the recessions of 2008–2009 and 2011–2012. The term premium shock χ_t increases during 2008–2009 and again during 2011, consistent with financial stress in the euro zone. As in [Bocola and Dovis \(2019\)](#), these two fundamental forces (the domestic recession and the tightening of global financial conditions) push the government closer to the region where confidence crises become possible and account for the fundamental component of the spread.

Figure 7: Fundamental Shocks During the Italian Crisis



Main result. Confidence risk accounts for approximately 80 percent of the Italian sovereign spread at its late-2011 peak (Figure 8, Panel a). Fundamental risk alone cannot account for the surge. The model also tracks the contemporaneous shortening of debt maturity (Panel b).

Figure 8: Italian Crisis



The filtered confidence states confirm this reading. The probability of a confidence crisis fluctuates at moderate levels through 2008–2011, then spikes to $\lambda \approx 0.08$, the grid maximum, at the late-2011/early-2012 peak. The filtered probability of the bad sunspot, $P(\omega_t = 1 \mid \text{data})$, stays near zero throughout, consistent with Italy never defaulting. The crisis is one of rising vulnerability rather than coordination on the bad equilibrium. Appendix D plots both filtered paths.

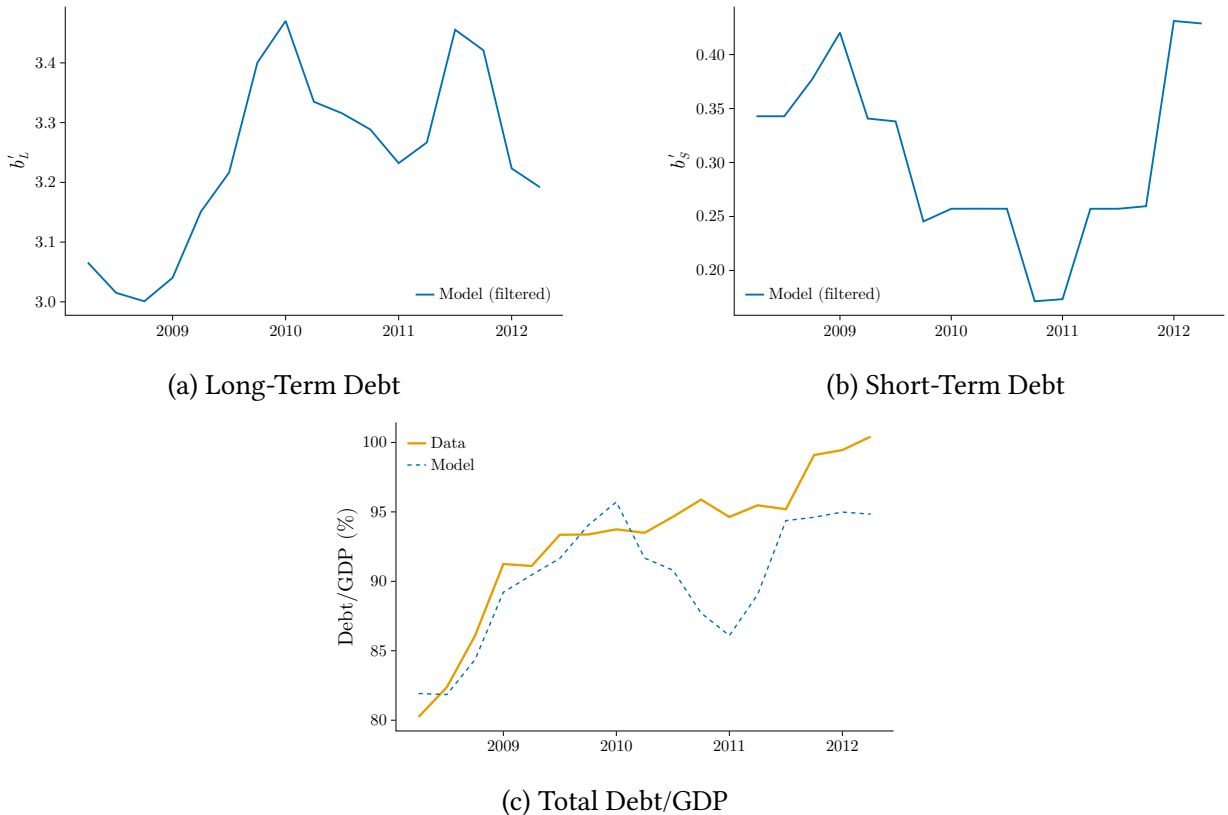
Comparison to Bocola–Dovis. Bocola and Dovis (2019) show that dropping maturity from their observables allows their model to attribute a larger share of the 2011 spread spike to rollover risk, but at the cost of predicting that the government should have *lengthened* its maturity by 0.5

years, the opposite of what we observed in the data. The tension arises because, in the Cole–Kehoe framework, higher rollover risk makes long-term debt the insurance instrument. In our model, the mechanism is reversed: higher λ makes short-term debt the insurance instrument, and the government responds by shortening maturity. Our model therefore simultaneously fits the spread surge and the maturity shortening without any tension between the two observables.

They also find $\omega \approx 0$ throughout the episode, but for a different reason and with different implications. In their Cole–Kehoe framework, a sunspot realization implies immediate default and full market exclusion. Since Italy never defaulted, the bad sunspot was never realized in their model either. As a result, confidence can affect spreads only through a forward-looking pricing of the probability that a future crisis *might* occur. In contrast, in our model the “bad” sunspot might trigger a confidence crisis with low prices for long-term bonds and no default. However, quantitatively, the filter finds a small role for this channel.

Debt dynamics. The filtered portfolio shows long-term debt rising through 2009–2010, short-term debt following a U-shape with a 2011–2012 surge, and total debt-to-GDP overshooting the data in 2009–2010 before converging (Figure 9).

Figure 9: Debt Dynamics



Panel (a) shows that long-term debt increases through 2009–2010 and then remains elevated before falling in late 2011. Panel (b) shows the U-shaped path for short-term debt: it falls from a local peak in 2009 to a trough around mid-2011 and then surges through the end of the sample. Panel (c) compares the total debt-to-GDP ratio in the model and the data. The model overshoots the data during 2009–2010 (reaching close to 100 percent of GDP against around 93 percent in the data), then converges to match the data through 2011 and 2012.

The rise in short-term issuance from 2011 to 2012 is consistent with the model’s insurance mechanism. As confidence risk (λ) peaked during the episode, the government optimally tilted its portfolio toward short-term debt to reduce its residual exposure on the long-term Laffer curve. The concurrent decline in long-term debt reflects the same rebalancing: as short-term financing expands, the budget needs less long-term issuance to clear.

Finally, we highlight that the particle filter recovers the debt portfolio from maturity and spreads, not from debt data. The model’s fit of the observed debt-to-GDP path through 2011–2012 (Panel c) is therefore an untargted moment.

Robustness. Appendix F reports robustness checks and a counterfactual that isolates the contribution of the maturity-shortening policy response to λ , a transmission channel absent in [Bocola and Dovis \(2019\)](#).

7 Conclusion

This paper revisits the quantitative importance of self-fulfilling risk in the European sovereign debt crisis. We find that shifts in investor beliefs account for approximately 80 percent of Italian sovereign spreads at the late-2011 peak, against the 13 percent that [Bocola and Dovis \(2019\)](#) attribute to confidence risk. The reversal comes from a Calvo-style mechanism in which short-term debt, not long-term debt, is the insurance instrument against price collapses; in this framework, a rising probability of crisis pushes the optimal maturity down rather than up. The same observed maturity shortening that signals low perceived risk in a Cole–Kehoe model signals high perceived risk in ours.

Governments facing fragile market confidence have an active policy lever in debt maturity management: they can shape their exposure to self-fulfilling crises by shortening maturity when confidence risk rises, not lengthening it. Extending this framework to monetary unions, where the central bank can affect the laffer curve directly, would allow a structural evaluation of interventions like the ECB’s 2012 announcement.

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A Proofs

This appendix extends the stylized analysis of Section ?? to the full model — stochastic y, χ, λ and endogenous Laffer-curve prices Q_L . In the stylized environment the analytical result follows from primitives because three of the underlying conditions hold by construction (concavity of u , exogenous and positive Q_S , the inequality $Q_L^b < Q_L^g$ that defines the two equilibria). In the full model these are no longer automatic: they become *state-by-state diagnostics* that identify the set of states at which short-term debt acts as insurance. We list these diagnostics, derive the comparative static $\partial b_S^*/\partial \lambda > 0$ under them, and state the analog diagnostic for the Cole–Kehoe region.

Setup. Fix state (s, λ) , so the fiscal rule pins down $g = G(b_S, b_L)$ and the state pins down tax revenue τY before the government acts. The government chooses b'_S before the sunspot $\omega \in \{0, 1\}$ is drawn, with $\Pr(\omega = 1) = \lambda$. After ω is realized, b'_L clears the budget at either the good equilibrium $b'_{L,g}(b'_S) \equiv \min \mathbb{B}_L^+(s, \lambda, b'_S)$ or the bad equilibrium $b'_{L,b}(b'_S) \equiv \max \mathbb{B}_L^+(s, \lambda, b'_S)$. The government's payoff is

$$W(\lambda; b'_S) = u(g) + \beta \left[(1 - \lambda) \mathcal{V}(b'_S, b'_{L,g}(b'_S)) + \lambda \mathcal{V}(b'_S, b'_{L,b}(b'_S)) \right], \quad (24)$$

where $\mathcal{V}(b'_S, b'_L) \equiv \mathbb{E}_{y', \chi', \lambda'} [V(b'_S, b'_L, y', \chi', \lambda')]$ is the next-period continuation value as a function of the inherited debt portfolio, integrating out the exogenous shocks at the model's equilibrium policy.

State-by-state diagnostics for the insurance channel. The following four quantities are computable at any state $(s, \lambda, b_S, b_L, b'_S)$ in the calibrated model:

- (i) Concavity of the continuation value in long-term debt: $\partial^2 \mathcal{V} / \partial (b'_L)^2 \leq 0$;
- (ii) Positive marginal revenue from short-term issuance: $\partial \mathcal{L} / \partial b'_S > 0$, where

$$\frac{\partial \mathcal{L}}{\partial b'_S} = Q_S + b'_S \frac{\partial Q_S}{\partial b'_S} + (b'_L - (1 - \delta)b_L) \frac{\partial Q_L}{\partial b'_S}; \quad (25)$$

- (iii) Slope asymmetry on the Laffer curve: $\mathcal{L}'(b'_{L,b}) < \mathcal{L}'(b'_{L,g})$ (the bad root sits on the flat part of the recovery branch);
- (iv) Dominance of the within-period insurance channel over the cross-period rollover channel: $\Delta_{\text{within}} + \Delta_{\text{cross}} \geq 0$, where

$$\begin{aligned} \Delta_{\text{within}} &\equiv \mathcal{V}_{b_L}(b'_S, b'_{L,b}) \frac{\partial b'_{L,b}}{\partial b'_S} - \mathcal{V}_{b_L}(b'_S, b'_{L,g}) \frac{\partial b'_{L,g}}{\partial b'_S}, \\ \Delta_{\text{cross}} &\equiv \mathcal{V}_{b_S}(b'_S, b'_{L,b}) - \mathcal{V}_{b_S}(b'_S, b'_{L,g}). \end{aligned}$$

Proposition 2 (Sufficient diagnostics for the insurance comparative static, full model). *Suppose the state (s, λ) lies in the multiplicity region $\mathcal{S}_M$ for the relevant range of b'_S , so that both equilibria coexist, and that diagnostics (i)–(iv) hold at that state. Then $\partial^2 W / \partial \lambda \partial b'_S > 0$, and the optimal short-term debt level $b_S^*(\lambda) \equiv \arg \max_{b'_S} W(\lambda; b'_S)$ is locally increasing in λ .*

In the stylized environment of Section ??, diagnostics (i)–(iii) hold by construction or reduce to the primitive parameter restriction (??) on $\beta, \kappa_L, \tilde{v}, \delta$. Only (iv) is a substantive restriction on the continuation value. In the full model with stochastic shocks and endogenous prices, none of (i)–(iv) hold by construction. Section 5 reports their numerical evaluation at every filtered Italian state.

Proof of Proposition 1. We first derive the stationary equilibrium prices in closed form. The short bond is auctioned at time t before ω_t is drawn and pays $1 + \kappa_S$ at $t + 1$ if the government does not default, with the per-unit recovery $\tilde{v}$ otherwise. In the stationary equilibrium of the stylized environment the bad sunspot $\omega = 1$ triggers default with the bond settling at the recovery floor, so the unconditional next-period default probability equals λ . Lenders therefore price the short bond at the expected discounted payoff,

$$Q_S = \beta[(1 - \lambda)(1 + \kappa_S) + \lambda \tilde{v}], \quad (26)$$

which gives (23). The bad-sunspot long-bond price satisfies (21); the good-sunspot pricing equation (22) is linear in Q_L^g and yields the closed form

$$Q_L^g(\lambda) = \frac{\beta(1 - \lambda)\kappa_L + \beta\lambda \tilde{v}}{1 - \beta(1 - \lambda)(1 - \delta)}. \quad (27)$$

Under (??), $Q_L^g(\lambda) > \tilde{v} = Q_L^b$ for every $\lambda \in [0, 1]$. The three prices Q_S, Q_L^g, Q_L^b are constants in b'_S at the stationary equilibrium: a marginal deviation in b'_S does not alter the stationary equilibrium at which they are evaluated. Differentiation of the period budget identity therefore gives $\partial b'_L(\omega) / \partial b'_S = -Q_S / Q_L^\omega$. The government's first-order condition at the stationary policy b_S^* is

$$(1 - \lambda) \left[\mathcal{V}_{b_S}(b_S^*, b_L^g) - \frac{Q_S}{Q_L^g} \mathcal{V}_{b_L}(b_S^*, b_L^g) \right] + \lambda \left[\mathcal{V}_{b_S}(b_S^*, b_L^b) - \frac{Q_S}{\tilde{v}} \mathcal{V}_{b_L}(b_S^*, b_L^b) \right] = 0. \quad (28)$$

Differentiating (28) with respect to λ at the optimum decomposes the cross-partial $\partial^2 W / (\partial \lambda \partial b'_S)$ into three terms. The first is the weight effect, $\left[\mathcal{V}_{b_S}(b_S^*, b_L^b) - \frac{Q_S}{\tilde{v}} \mathcal{V}_{b_L}(b_S^*, b_L^b) \right] - \left[\mathcal{V}_{b_S}(b_S^*, b_L^g) - \frac{Q_S}{Q_L^g} \mathcal{V}_{b_L}(b_S^*, b_L^g) \right]$, which captures the difference in marginal benefit across the two sunspot states; under the price asymmetry (??) and $\mathcal{V}_{b_L} < 0$ (Lemma 1), the term $-Q_S \mathcal{V}_{b_L} / Q_L^\omega$ is larger at the bad sunspot, so the weight effect is positive. The second is the price effect: differentiating (27) yields

$$\frac{\partial Q_L^g}{\partial \lambda} = \frac{\beta(\tilde{v} - \kappa_L) [1 - \beta(1 - \lambda)(1 - \delta)] - \beta(1 - \delta) [\beta(1 - \lambda)\kappa_L + \beta\lambda \tilde{v}]}{[1 - \beta(1 - \lambda)(1 - \delta)]^2} < 0$$

under (??), so $1/Q_L^g$ is increasing in λ and the magnitude of the good-state insurance term $-Q_S \mathcal{V}_{b_L}(b_S^*, b_L^g)/Q_L^g$ rises with λ . This effect is strictly positive. The third is the cross-period effect, $\mathcal{V}_{b_S}(b_S^*, b_L^b) - \mathcal{V}_{b_S}(b_S^*, b_L^g)$, whose sign is governed by the cross-partial $\mathcal{V}_{b_S b_L}$; under $\mathcal{V}_{b_S b_L} \leq 0$ and $b_L^b > b_L^g$, this effect is non-positive. Condition (iv) bounds the magnitude of the cross-period effect, so the sum of the three effects is strictly positive, $\partial^2 W / (\partial \lambda \partial b_S') > 0$. Topkis's theorem delivers $\partial b_S^* / \partial \lambda > 0$.

We next establish a property of the continuation value used in the proof of the full-model proposition.

Lemma 1 (Monotonicity of the continuation value). *Under the paper's primitives (CRRA utility, fiscal rule $G(b_S, b_L) = \bar{g} - \alpha_1 \max(b_S + b_L - \bar{b}, 0)$, default rule with autarky spending $g_D < \tau Y_{\min}$, and pari-passu recovery $\tilde{v} = \phi(\tau Y' - g_D)/(b_S' + b_L')$), the continuation value satisfies $\mathcal{V}_{b_L}(b_S', b_L') < 0$ for all $b_L' > 0$.*

Proof. The continuation value $\mathcal{V}(b_S', b_L')$ is the expected next-period-and-beyond welfare from inheriting the portfolio (b_S', b_L') , holding all other states fixed and integrating out the exogenous shocks at the model's equilibrium policy. Raising b_L' enters the next-period problem through three channels, each of which weakly reduces welfare:

(a) *Debt service.* The coupon payment $\kappa_L b_L'$ scales linearly with b_L' , tightening the next-period budget constraint (5). Holding all else fixed, this requires either lower g' or higher new issuance, both of which reduce welfare.

(b) *Fiscal rule.* When total inherited debt exceeds $\bar{b}$, the fiscal rule reduces g' at rate α_1 in b_L' . Since $u'(g) = g^{-\sigma} > 0$ under CRRA, $u(g')$ strictly falls. This channel is silent for $b_S' + b_L' \leq \bar{b}$ but does not change sign there.

(c) *Default probability.* Higher b_L' enters the period- $t + 1$ deficit linearly through the coupon obligation $\kappa_L b_L'$, shrinking the set of states $(Y_{t+1}, \chi_{t+1}, \lambda_{t+1})$ for which the Laffer curve admits a stable budget-clearing root. The probability of default $\Pr(d' = 1)$ is therefore weakly increasing in b_L' . Default delivers period payoff $u(g_D) < u(\tau Y') \leq u(g')$, since $g_D < \tau Y_{\min}$ by construction; the per-period payoff under default is strictly less than the per-period payoff in any non-default state. The expected per-period utility is therefore strictly decreasing in b_L' through the default-probability channel, independent of any comparison between V_{def} and the no-default continuation value.

At any $b_L' > 0$, channels (a) and (c) operate strictly, so $\mathcal{V}_{b_L}(b_S', b_L') < 0$ holds at any state where the no-default continuation does not over-compensate for the period-payoff drop via the debt-discharge effect. Section 5 verifies this inequality, together with the four conditions of Proposition 2, at every filtered Italian quarter. $\square$

Proof of Proposition 2. Since both roots lie on upward-sloping portions of $\mathcal{L}$ (stable equilibria), $\mathcal{L}'(b_{L,\omega}') > 0$ for $\omega \in \{g, b\}$. The implicit function theorem on $\mathcal{L}(s, \lambda, b_S', b_{L,\omega}') = g - \tau Y$ gives

$$\frac{\partial b_{L,\omega}'}{\partial b_S'} = -\frac{\partial \mathcal{L} / \partial b_S'}{\mathcal{L}'(b_{L,\omega}')} < 0,$$

where the sign uses condition (ii). By condition (iii), $\mathcal{L}'(b'_{L,b}) < \mathcal{L}'(b'_{L,g})$, so

$$\left| \frac{\partial b'_{L,b}}{\partial b'_S} \right| > \left| \frac{\partial b'_{L,g}}{\partial b'_S} \right| :$$

a marginal increase in b'_S reduces the bad-equilibrium debt stock by more than the good-equilibrium stock.

Differentiating $W(\lambda; b'_S) = u(g) + \beta[(1-\lambda)\mathcal{V}(b'_S, b'_{L,g}(b'_S)) + \lambda\mathcal{V}(b'_S, b'_{L,b}(b'_S))]$ with respect to b'_S gives

$$\begin{aligned} \frac{\partial W}{\partial b'_S} = & \beta \left\{ (1-\lambda) \left[\mathcal{V}_{b_S}(b'_S, b'_{L,g}) + \mathcal{V}_{b_L}(b'_S, b'_{L,g}) \frac{\partial b'_{L,g}}{\partial b'_S} \right] \right. \\ & \left. + \lambda \left[\mathcal{V}_{b_S}(b'_S, b'_{L,b}) + \mathcal{V}_{b_L}(b'_S, b'_{L,b}) \frac{\partial b'_{L,b}}{\partial b'_S} \right] \right\}. \end{aligned}$$

Taking the derivative with respect to λ and collecting terms,

$$\frac{\partial^2 W}{\partial \lambda \partial b'_S} = \beta(\Delta_{\text{within}} + \Delta_{\text{cross}}),$$

with

$$\Delta_{\text{within}} = \mathcal{V}_{b_L}(b'_S, b'_{L,b}) \frac{\partial b'_{L,b}}{\partial b'_S} - \mathcal{V}_{b_L}(b'_S, b'_{L,g}) \frac{\partial b'_{L,g}}{\partial b'_S}, \quad \Delta_{\text{cross}} = \mathcal{V}_{b_S}(b'_S, b'_{L,b}) - \mathcal{V}_{b_S}(b'_S, b'_{L,g}).$$

The two pieces correspond to the two channels through which a marginal b'_S affects welfare differently in the bad versus good equilibrium: Δ_{within} captures the within-period insurance channel (the implicit-function chain through b'_L), and Δ_{cross} captures the cross-period rollover channel (the direct dependence of $\mathcal{V}$ on inherited short-term debt).

Signing Δ_{within} . Substituting the implicit-function expressions,

$$\Delta_{\text{within}} = \frac{\partial \mathcal{L}}{\partial b'_S} \left[\frac{-\mathcal{V}_{b_L}(b'_S, b'_{L,b})}{\mathcal{L}'(b'_{L,b})} - \frac{-\mathcal{V}_{b_L}(b'_S, b'_{L,g})}{\mathcal{L}'(b'_{L,g})} \right].$$

The prefactor is positive by condition (ii). By Lemma 1, $-\mathcal{V}_{b_L}(b'_S, b'_{L,\omega}) > 0$. Since $b'_{L,b} > b'_{L,g}$, condition (i) (concavity in b_L) implies $-\mathcal{V}_{b_L}(b'_S, b'_{L,b}) \geq -\mathcal{V}_{b_L}(b'_S, b'_{L,g})$. Combined with $1/\mathcal{L}'(b'_{L,b}) > 1/\mathcal{L}'(b'_{L,g})$ from condition (iii), the bracketed expression is non-negative, so $\Delta_{\text{within}} \geq 0$ with strict inequality under strict versions of (i)–(iii).

Total cross-partial. Condition (iv) states $\Delta_{\text{within}} + \Delta_{\text{cross}} \geq 0$, with strict inequality under strict (i)–(iv). Hence $\partial^2 W / (\partial \lambda \partial b'_S) > 0$, and Topkis's theorem implies $\partial b_S^*(\lambda) / \partial \lambda > 0$. $\square$

Remark 1 (Sign of the cross-period channel). *The direct cross-period term Δ_{cross} depends on $\mathcal{V}_{b_S b_L}$, the cross-partial of the continuation value in (b_S, b_L) . In sovereign-default models with non-trivial default risk, more inherited long-term debt typically amplifies the marginal cost of inherited short-*

term debt, so $\mathcal{V}_{b_S b_L} \leq 0$ and hence $\Delta_{\text{cross}} \leq 0$. The cross-period channel therefore partially offsets the within-period insurance channel. Condition (iv) requires the within-period channel to dominate; Section 5 verifies numerically that this dominance holds over the calibrated parameter region.

Verification of Proposition 1 on the filtered path. We evaluate the four sufficient conditions of Proposition 2 (the full-model statement) in each filtered Italian quarter (2008:Q2–2012:Q2). Condition (i) holds strictly at every quarter: the second finite difference of $\mathcal{V}$ in b'_L is strictly negative, ranging from -1.4 to -2.9 . Condition (ii) holds strictly: $\partial \mathcal{L} / \partial b'_S$ lies in $[0.90, 1.00]$, close to its theoretical upper bound of $1 + \kappa_S$. Conditions (iii) and (iv) hold for every particle for which the underlying state lies in the multiplicity region. The bad-sunspot realization $\omega = 1$ is rare in posterior mass (essentially zero in every quarter), consistent with Italy not defaulting on the sample.

B Computational Algorithm

We solve the model by value function iteration on the discretized state space $(b_S, b_L, y, \chi, \lambda)$, combining grid-based Laffer curve classification with logit-smoothed policy functions following [Dvorkin et al. \(2021\)](#).

Grids and Initialization

Output y and the SDF state χ are each discretized by Tauchen (1986) and combined into a joint Markov chain via Gaussian quadrature, producing transition matrix $\Pi_{y\chi}$ and conditional SDF weights $\{M_{ij}\}$. The confidence shock λ is discretized on a uniform grid over $[\lambda_{\min}, \lambda_{\max}]$ with an AR(1) Tauchen-style transition matrix Π_λ . A fine evaluation grid $b_{L,\text{eval}} \in [0, \bar{b}_L]$ of 500 points is used only for Laffer curve evaluation; equilibrium debt levels are stored and interpolated on the coarser b_L state grid.

The value function is initialized at the no-default steady state, $V^0 = u(\bar{g})/(1 - \beta)$. Bond prices use a recovery-based initialization, $Q_L^0(b'_L) = \min(Q_L^{\text{rf}}, \phi(\tau Y - g_D)/b'_L)$. This seeds the declining $Q_L(b'_L)$ profile needed for the Laffer curve to exhibit a second hump from the first iteration. Short-term prices are initialized at their risk-free values, $Q_S^0 = Q_S^{\text{rf}}$.

Pre-Logit Value and Choice Probabilities

For each state $(s, \lambda) = (b_S, b_L, y, \chi, \lambda)$ and candidate b'_S , define

$$W(s, \lambda, b'_S) = (1 - \lambda) [u(\tilde{g}_0) + \beta \mathbb{E}_{y', \chi', \lambda'} V(s'_g, \lambda')] + \lambda [u(\tilde{g}_1) + \beta \mathbb{E}_{y', \chi', \lambda'} V(s'_b, \lambda')], \quad (29)$$

where $\tilde{g}_\omega = g$ if the ω -branch repays and $\tilde{g}_\omega = g_D$ if it defaults, and $s'_g = (b'_S, b_{L,g}^*, y', \chi')$ and $s'_b = (b'_S, b_{L,b}^*, y', \chi')$ are next-period states under each sunspot. Choice probabilities and the value function are computed using a numerically stable logit, where $\rho > 0$ is the taste-shock variance:

$$\Pr(b'_S = b'_{S,k} \mid s, \lambda) = \frac{\exp((W_k - \bar{W})/\rho)}{\sum_j \exp((W_j - \bar{W})/\rho)}, \quad V(s, \lambda) = \bar{W} + \rho \log \sum_j \exp\left(\frac{W_j - \bar{W}}{\rho}\right), \quad (30)$$

where $\bar{W} = \max_j W_j$.

Algorithm

Each outer iteration proceeds as follows.

1. **Laffer curve classification.** For each state $(b_S, b_L, y, \chi, \lambda)$ and each candidate b'_S , compute the financing need $\Delta = g - \tau Y + (1 + \kappa_S)b_S + \kappa_L b_L$ and evaluate

$$\mathcal{L}(b'_L) = Q_L(y, \chi, \lambda, b'_S, b'_L) \cdot (b'_L - (1 - \delta)b_L) + \bar{Q}_S \cdot b'_S$$

on the fine grid, where $\bar{Q}_S = (1 - \lambda)Q_S(b_{L,g}^*) + \lambda Q_S(b_{L,b}^*)$ is the previous-iteration equilibrium-weighted short-term price (since lenders price Q_S before the sunspot is drawn). Identify stable crossings where $\mathcal{L}$ crosses Δ from below. Cases 2, 3, 4 below correspond to $\mathcal{S}_U, \mathcal{S}_{CK}, \mathcal{S}_M$ in Section 4; Case 1 (no equilibrium) is captured by the default rule (16).

- **Case 1 (Fundamental default):** No stable crossings: both sunspots default.
 - **Case 2 (Unique equilibrium):** One stable crossing and $\mathcal{L}(\bar{b}_L) \geq \Delta$: both sunspots select the same root; ω is irrelevant.
 - **Case 3 (Cole–Kehoe):** One stable crossing and $\mathcal{L}(\bar{b}_L) < \Delta$: good sunspot repays at the unique root; bad sunspot defaults.
 - **Case 4 (LW multiplicity):** Two or more stable crossings and $\mathcal{L}(\bar{b}_L) \geq \Delta$: good sunspot selects the lowest root ($b_{L,g}^*$), bad sunspot selects the highest ($b_{L,b}^*$).
2. **Value function update.** Compute $W(s, \lambda, b'_S)$ for all b'_S candidates, then update V and $\Pr(b'_S)$ via (30).
 3. **Price updates.** Update Q_L and Q_S from the no-arbitrage conditions (19)–(20), integrating over the current logit probabilities. For Q_L , we evaluate continuation prices at the previous-iteration Q_L following Chatterjee and Eyigungor (2012).
 4. **Damped accumulator.** A damped accumulator Q^{main} is updated each iteration via $Q^{\text{main}} \leftarrow (1 - \epsilon)Q^{\text{main}} + \epsilon Q^{\text{new}}$ with damping factor $\epsilon = 0.05$, following Chatterjee and Eyigungor (2012). Convergence is checked on Q^{main} : the algorithm stops when $\|Q_L^{\text{main}} - Q_L^{\text{prev}}\|_\infty, \|Q_S^{\text{main}} - Q_S^{\text{prev}}\|_\infty$, and $\|V^{\text{main}} - V^{\text{prev}}\|_\infty$ all fall below 10^{-5} .

C Empirical Definitions

Maturity. The portfolio maturity is a face-value-weighted average of the maturities of the two instruments. Short-term bonds have maturity 1 (one quarter). Long-term bonds have risk-free Macaulay duration $1/\delta$ (quarters). We use risk-free duration to isolate changes in quantities from movements in bond prices. Dividing by 4 converts to annual years:

$$\text{mat}_t = \frac{b'_S \cdot 1 + b'_L \cdot (1/\delta)}{4(b'_S + b'_L)}.$$

Spreads. The short-term spread is the excess gross return of the sovereign bond over the risk-free short rate:

$$s_S = 400 \times \left(\frac{1 + \kappa_S}{q_S} - R_{f,S} \right),$$

where $(1 + \kappa_S)/q_S$ is the sovereign gross return and $R_{f,S}$ is the state-dependent risk-free gross return. The long-term spread is the excess yield-to-maturity over the risk-free long yield:

$$s_L = 400 \times (\text{YTM}_L - R_{f,L}), \quad \text{YTM}_L = \frac{\kappa_L}{q_L} - \delta.$$

Both are in annualized percentage points (multiply by 100 to convert to basis points). The portfolio spread is a market-value-weighted average:

$$s = \frac{q_S b'_S \cdot s_S + q_L b'_L \cdot s_L}{q_S b'_S + q_L b'_L}.$$

D Particle Filter

We cast the model as a nonlinear state-space system and estimate the unobserved states using a bootstrap Sequential Monte Carlo (particle) filter. This appendix describes the state vector, measurement equation, and algorithm.

State Vector and Observables

The state vector is $S_t = (y_t, \chi_t, \lambda_t, \omega_t, b_{S,t}, b_{L,t})$. The fundamentals (y_t, χ_t) are observed directly from the data and assigned zero measurement error, so filtered values coincide exactly with the data series. The confidence shocks (λ_t, ω_t) and the debt stocks $(b_{S,t}, b_{L,t})$ are latent and recovered by the filter.

The observables are portfolio maturity and the sovereign spread. The measurement equation is

$$\begin{pmatrix} \text{maturity}_t \\ \text{spread}_t \end{pmatrix} = \begin{pmatrix} \hat{m}(S_t) \\ \hat{s}(S_t) \end{pmatrix} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, H^{-1}),$$

where $\hat{m}$ and $\hat{s}$ are the model-implied maturity and spread evaluated at state S_t via interpolation of the solved policy and price functions, and H is the 2×2 precision matrix for measurement errors. We set the measurement error standard deviations to $\sigma_{\text{maturity}} = 0.05$ years and $\sigma_{\text{spread}} = 0.08$ (in model units, where $1.0 = 100$ bps), with zero correlation.

Initialization

We initialize $N = 50,000$ particles at $t = 1$. The initial debt portfolio $(b_{S,0}, b_{L,0})$ is recovered from the first observation using observed maturity (m_0) and total debt-to-GDP (d_0), exploiting the identity

$$m_0 = \frac{b_{S,0} \cdot 1 + b_{L,0} \cdot (1/\delta)}{b_{S,0} + b_{L,0}}, \quad b_{S,0} + b_{L,0} = d_0 / (4 \cdot e^{y_0}).$$

Each particle draws $b_{S,0}^k \sim \mathcal{N}(b_{S,0}, 0.05^2)$ and $b_{L,0}^k \sim \mathcal{N}(b_{L,0}, 0.10^2)$, clamped to the grid bounds. The confidence shock is initialized as $\lambda_0^k \sim \text{Uniform}[\lambda_{\min}, \lambda_{\max}]$ and $\omega_0^k \sim \text{Bernoulli}(0.5)$.

Algorithm

For each period $t = 1, \dots, T$:

1. **Set fundamentals.** Set $y_t^k = y_t$ and $\chi_t^k = \chi_t$ for all particles, taken directly from the data.
2. **Propagate confidence shocks.** For each particle k , draw λ_t^k from the Markov transition Π_λ using the previous period's resampled λ_{t-1}^k as the current state. Then draw $\omega_t^k \sim \text{Bernoulli}(\lambda_t^k)$.

3. **Evaluate policies and prices.** For each particle, interpolate the government's short-term debt choice $b_{S,t}^{\prime k}$ from the logit choice distribution at state S_t^k , and the corresponding long-term debt $b_{L,t}^{\prime k}$ from the equilibrium debt rule conditional on ω_t^k . Compute model-implied maturity $\hat{m}_t^k$ and spread $\hat{s}_t^k$ using interpolated price functions.
4. **Compute weights.** Particles in default receive weight zero. For repaying particles, the unnormalized weight is the Gaussian likelihood

$$\tilde{w}_t^k = \exp\left(-\frac{1}{2} \begin{pmatrix} \hat{m}_t^k - m_t \\ \hat{s}_t^k - s_t \end{pmatrix}^\top H \begin{pmatrix} \hat{m}_t^k - m_t \\ \hat{s}_t^k - s_t \end{pmatrix}\right).$$

Normalize to obtain $w_t^k = \tilde{w}_t^k / \sum_j \tilde{w}_t^j$. If the effective sample size $\text{ESS}_t = 1 / \sum_k (w_t^k)^2$ falls below $0.01 N$, we replace the likelihood weights with an inverse-distance scheme to prevent particle collapse. Specifically, let $\Delta_t^k = (\hat{m}_t^k - m_t, \hat{s}_t^k - s_t)^\top$ denote the period- t observation residual for particle k , and define the Mahalanobis distance

$$d_t^k \equiv (\Delta_t^k)^\top H \Delta_t^k,$$

using the same precision matrix H as in the likelihood. The fallback weights are

$$\tilde{w}_t^k = (d_t^k + \varepsilon)^{-\zeta},$$

with $\zeta = 5.5$ and $\varepsilon = 10^{-9}$ for numerical regularization, then normalized to sum to one. The fallback fires only in the rare quarters where the likelihood collapses below the $0.01 N$ ESS threshold (typically 1–2 quarters at the height of the crisis); in all other periods the standard likelihood weighting applies.

5. **Resample.** Apply systematic resampling to obtain N equally weighted particles. Carry forward the resampled λ_t^k for the Markov transition in the next period.
6. **Propagate debt.** Set $b_{S,t+1}^k = b_{S,t}^{\prime k}$ and $b_{L,t+1}^k = b_{L,t}^{\prime k}$ for the next period. Defaulted particles re-enter with debt at the grid minimum.

Spread Decomposition

At each particle and period, we compute three model-implied spreads corresponding to the decomposition in Section 6. All three use the same realized state $(y_t, \chi_t, b_{S,t}, b_{L,t})$ and the same $b_{S,t}^{\prime}$ draw, so differences reflect only the pricing of confidence risk.

- **Fundamental spread:** prices evaluated at $\lambda = \lambda_{\min}$ and $\omega = 0$, shutting down both confidence margins.

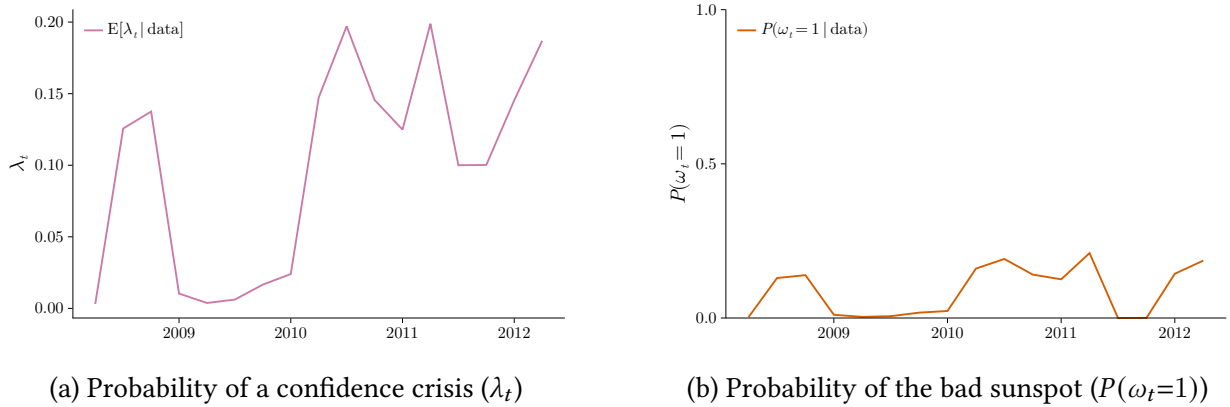
- **Fundamental + confidence spread:** prices evaluated at the filtered λ_t^k but with $\omega = 0$. This is the cumulative level combining both the fundamental and confidence-risk components; the confidence-risk component alone is the difference between this spread and the fundamental spread.
- **Realized spread:** prices evaluated at the realized $(\lambda_t^k, \omega_t^k)$, matching the full model prediction.

We compute period- t aggregate values as weighted averages $\sum_k w_t^k \cdot (\cdot)^k$ across particles.

Filtered Confidence States

Figure 10 plots the two filtered confidence states referenced in Section 6. The probability of a confidence crisis λ_t rises into the late-2011 peak. The probability of the bad sunspot $P(\omega_t = 1 \mid \text{data})$ stays near zero throughout, the filter's reading that Italy never coordinated on the bad equilibrium.

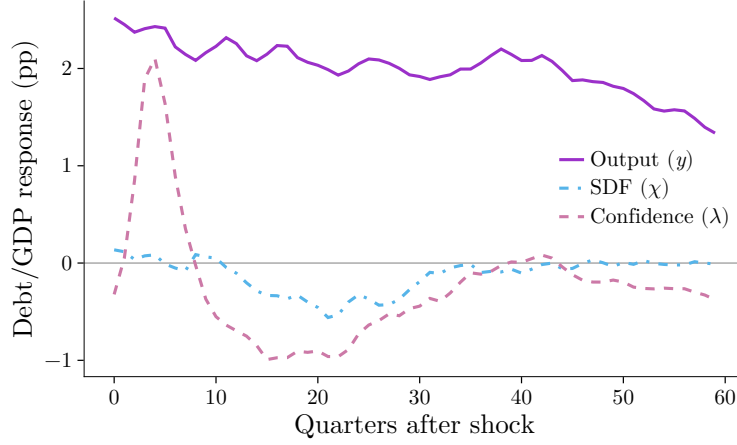
Figure 10: Filtered Confidence States



E Debt Impulse Responses

Figure 11 reports the debt-to-GDP response to the three shocks, completing the impulse responses of Section 5. A confidence shock raises debt for about a year as the government rolls over its larger short-term position, after which the response unwinds. The output response is persistent because debt-to-GDP is a slow-moving stock that mean-reverts only gradually.

Figure 11: Debt/GDP Impulse Responses



F Sensitivity and Robustness

This appendix collects four diagnostics that probe the central quantitative results of Section 6. First, the bad-equilibrium intersection in Figure 3 sits on a locally upward-sloping segment of the Laffer curve (slope $+0.009$ at the bad root vs $+0.914$ at the good root), confirming the stability and slope asymmetry of Proposition 1. Second, the posterior 5/95 band on the model spread is tight at most dates and widest at the late-2011 peak (345–474 bps at 2011:Q4). Third, the headline filtered series are stable across particle count and random seed but sensitive to the measurement-error variances in the H matrix. Fourth, we report an alternative fundamental counterfactual in which the government’s portfolio is re-optimized at $\lambda = \lambda_{\min}$ rather than held fixed at the realized policy. This alternative essentially attributes the entire spread to the confidence channel; the gap between it and the headline counterfactual measures the contribution of the maturity-shortening policy response to λ , a margin not present in Bocola and Dovis (2019).

Bad-equilibrium slope on the calibrated Laffer curve

Proposition 2 condition (iii) requires the Laffer curve to be flatter at the bad equilibrium than at the good equilibrium, with both intersections on locally increasing segments. Evaluating the curve plotted in Figure 3 at the low- b'_S value where the multiplicity occurs ($b'_S = 0.273$, Case 4), we find $\mathcal{L}'(b'_{L,g}) = +0.914$ at the good root and $\mathcal{L}'(b'_{L,b}) = +0.009$ at the bad root. Both slopes are strictly positive (stable equilibria), and the bad-root slope is two orders of magnitude smaller (the slope asymmetry of Proposition 2 (iii)).

Posterior uncertainty on the model spread

Figure 12 reports the weighted 5th, 50th, and 95th percentiles of the model spread across the particle swarm at each filtered quarter. The band is tight at most dates and widest at the late-2011 peak, ranging from 345 to 474 bps at 2011:Q4 against an observed value of 443 bps.

Figure 12: Posterior 5/95 bands on the model spread



Notes: Weighted 5th, 50th, and 95th percentiles of the model-spread distribution across the particle swarm at each filtered quarter; $N = 50,000$ particles. Shaded region is the 5/95 band; solid line is the median. The observed Italian spread is overlaid in amber.

Filter robustness to particle count and measurement error

Table 3 reports three summary statistics under perturbations to the particle count N , the maturity and spread measurement-error standard deviations σ_{mat} and σ_{spr} , and the random seed. The peak filtered λ , the average $\bar{\omega}$, and the peak model spread are all stable across particle counts (20,000 to 100,000) and seeds. Doubling either measurement-error variance moves the peak model spread by ± 100 bps, identifying the H matrix as the most consequential filter choice.

Table 3: Filter sensitivity

Configuration	λ_{peak}	$\bar{\omega}$	spread _{peak} (bps)
Baseline ($N = 50,000$, $\sigma_{\text{mat}} = 0.04$, $\sigma_{\text{spr}} = 0.08$)	0.054	0.026	269
$N = 20,000$	0.055	0.026	265
$N = 100,000$	0.061	0.023	262
σ_{mat} doubled	0.054	0.022	363
σ_{spr} doubled	0.050	0.030	164
Alternative random seed	0.054	0.026	273

Alternative fundamental counterfactual: re-optimized policy at $\lambda = \lambda_{\min}$

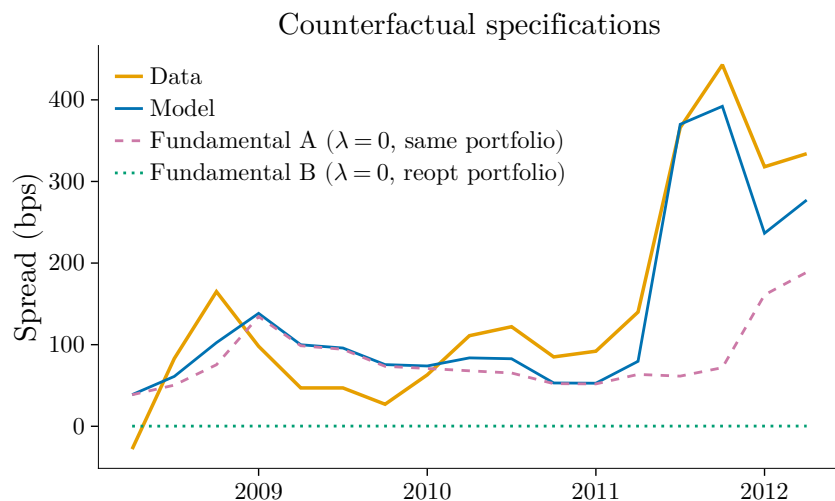
The fundamental component reported in Section 6 follows [Bocola and DAVIS \(2019\)](#): it holds the realized portfolio (b'_S, b'_L) fixed and re-prices the bond at $\lambda = \lambda_{\min}$. In their setting this construction isolates the pricing channel of confidence risk because the government has no portfolio margin to respond to λ . In our setting the government does respond by shortening maturity, so the realized portfolio embeds a confidence-driven policy choice. The Bocola–Davis construction therefore attributes part of the policy-response effect to the fundamental component.

Figure 13 compares two specifications:

- **Specification A** (the headline fundamental in Section 6): same realized portfolio, priced at $\lambda_{\min}$.
- **Specification B**: re-optimized portfolio at $\lambda_{\min}$ (the government chooses b'_S as if confidence risk were absent), priced at $\lambda_{\min}$.

Specification B averages 0.3 bps over the sample and reaches 0.3 bps at the late-2011 peak, against 96 bps and 195 bps respectively under Specification A. The full gap between the model spread and Specification B — essentially the entire spread, including the part Specification A labels fundamental — is attributable to the joint pricing and policy responses to confidence risk. The gap between Specifications A and B isolates the portion driven by the maturity-shortening policy response. Within our model, this is a substantive contribution of the framework over [Bocola and DAVIS \(2019\)](#): the maturity instrument is itself a transmission channel of λ , not merely a passive observable used to identify it.

Figure 13: Counterfactual specifications for the fundamental component



Notes: Data (amber), realized model spread (blue), Specification A (reddish-purple dashed; same portfolio, priced at $\lambda_{\min}$), Specification B (green dotted; re-optimized portfolio at $\lambda_{\min}$, priced at $\lambda_{\min}$). All quantities in basis points.